

HCIR for Swiss Open Government Data Using Fuzzy Logic-Based Ranking

Master's Thesis

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Abstract

Open Government Data (OGD) initiatives have significantly increased the availability of publicly accessible datasets, supporting transparency, innovation, and evidence-based decision-making. Despite these benefits, users frequently encounter difficulties when attempting to discover relevant datasets within large and heterogeneous OGD repositories. Traditional retrieval approaches primarily rely on keyword-based matching and often provide limited support for exploratory search, transparency, and explainability.

This thesis proposes a HCIR framework for Swiss OGD that combines BM25 retrieval, fuzzy logic-based relevance estimation, and explainable ranking mechanisms. The framework integrates multiple metadata-driven relevance factors, including keyword similarity, metadata completeness, and dataset recency, and resource availability through a fuzzy inference system designed to model gradual relevance and support human-centered interaction.

The research follows a DSR methodology involving problem identification, artifact design, implementation, and evaluation. A prototype was developed using metadata obtained from the Swiss OGD portal and evaluated through a user-centered study. The evaluation examined retrieval effectiveness, usability, transparency, explainability, and user trust.

The findings indicate that the proposed framework successfully supports dataset discoverability while improving transparency and user understanding of ranking decisions. Furthermore, the results demonstrate that fuzzy logic contributes meaningfully to HCIR by enabling gradual relevance estimation and interpretable decision-making. The study contributes to the fields of Information Retrieval, HCIR, and OGD by demonstrating that fuzzy logic can effectively complement conventional BM25 retrieval by providing transparent multi-criteria ranking while maintaining competitive retrieval effectiveness and improving the explainability of ranking decisions.

Keywords: Open Government Data, HCIR, Fuzzy Logic, BM25, Explainable Information Retrieval(XIR), Dataset Discovery, Information Retrieval.

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List of Abbreviations

API	Application Programming Interface
BM25	Best Matching 25
CKAN	Comprehensive Knowledge Archive Network
DCAT	Data Catalog Vocabulary
DSR	Design Science Research
HCIR	HCIR
HCI	Human-Computer Interaction
IR	Information Retrieval
NLP	Natural Language Processing
OGD	Open Government Data
SBERT	Sentence-BERT
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface
XAI	Explainable Artificial Intelligence
XIR	Explainable Information Retrieval

Chapter 1

Introduction

The rapid growth of OGD has increased opportunities for transparency, innovation, and evidence-based decision-making. Governments worldwide have made large volumes of public-sector data openly accessible through national data portals, enabling researchers, businesses, policymakers, and citizens to access and reuse information across domains such as transportation, health, education, environment, and public administration. As these repositories continue to expand, however, discovering relevant datasets has become increasingly difficult. Traditional keyword-based retrieval systems often struggle to satisfy complex or exploratory information needs, highlighting the need for retrieval approaches that better support human search behaviour.

This chapter introduces the research problem addressed in this thesis, identifies the research gap, presents the objectives and research questions, summarizes the main contributions, and outlines the structure of the thesis.

1.1 Background

OGD initiatives have become an important component of digital government strategies worldwide. By making public-sector data openly available, governments aim to improve transparency, accountability, citizen participation, and innovation [16, 1]. Open data enables researchers, businesses, policymakers, and citizens to access and reuse governmental information for scientific research, policy analysis, economic development, and public service innovation.

Many countries have established national open data portals that aggregate datasets published by public institutions. These portals provide centralized access to datasets originating from multiple governmental organizations and domains, including transportation, health, education, environment, energy, public administration, and demographics [1]. The availability of such datasets creates significant opportunities for evidence-based decision-making and data-driven innovation.

Switzerland actively supports OGD initiatives through the national portal *open-data.swiss*, which serves as a centralized platform for accessing datasets published by federal, cantonal, and municipal organizations [11]. The portal currently hosts thousands of datasets covering a wide range of thematic domains and provides access through both web interfaces and application programming interfaces (APIs).

Despite the increasing availability of public datasets, effective dataset discovery remains a significant challenge. As the number of available datasets grows, users often struggle to identify datasets that satisfy their information needs. The challenge is not only related to the quantity of available data but also to the complexity of metadata descriptions, inconsistencies across publishers, varying levels of metadata quality, and differences in terminology used to describe similar concepts [1].

Traditional dataset retrieval systems primarily rely on keyword-based search mechanisms. These systems typically rank datasets based on textual similarity between user queries and dataset metadata. Although such approaches are computationally efficient and widely adopted, they often fail to capture the complexity of human information needs. Users frequently express vague, exploratory, or partially defined queries that cannot be effectively interpreted using strict keyword matching alone [26].

This challenge becomes particularly important within the field of HCIR (HCIR), which emphasizes the interaction between users and retrieval systems. Rather than focusing solely on algorithmic ranking performance, HCIR seeks to design retrieval systems that align with human reasoning, support exploratory search behavior, and provide understandable explanations for retrieval decisions [26].

In this context, fuzzy logic presents a promising approach for modeling uncertainty and gradual relevance. Unlike traditional binary retrieval models that classify documents as either relevant or non-relevant, fuzzy logic enables relevance to be represented as a continuum. This capability makes fuzzy systems particularly suitable for handling vague concepts and supporting human-centered retrieval scenarios where relevance is inherently subjective and context-dependent [42, 22].

1.2 Problem Statement

The rapid growth of OGD repositories has significantly increased the complexity of dataset discovery. Although large quantities of data are available through platforms such as *opendata.swiss*, users frequently encounter difficulties when searching for relevant datasets.

Existing retrieval approaches primarily rely on lexical relevance and provide limited support for transparent multi-criteria ranking and exploratory search. These approaches estimate relevance based on statistical relationships between query terms and document metadata [35, 34]. While effective for exact-match retrieval scenarios, they often exhibit

several limitations when applied to exploratory dataset search.

First, traditional retrieval models are highly sensitive to vocabulary mismatch. Users may describe an information need using terminology that differs from the terminology used within dataset metadata. As a result, potentially relevant datasets may not appear among the top-ranked results.

Second, conventional ranking systems typically focus on textual similarity while neglecting other important metadata characteristics. Factors such as dataset recency, metadata completeness, update frequency, and thematic relevance may significantly influence dataset usefulness but are often not incorporated directly into ranking decisions.

Third, many retrieval systems operate as black-box mechanisms that provide little transparency regarding how relevance scores are computed. Users may receive ranked results without understanding why certain datasets were recommended or why others were excluded. This lack of transparency can reduce trust in retrieval systems and hinder effective decision-making [9].

Finally, human information needs are often vague, incomplete, and evolving. Users exploring OGD frequently begin with broad topics rather than precise dataset requirements. Traditional retrieval systems are not well equipped to support such exploratory search behavior because they assume clearly defined queries and exact matching conditions [26].

These limitations indicate the need for retrieval approaches that can integrate multiple relevance criteria, accommodate uncertainty, and provide transparent explanations for ranking decisions.

1.3 Research Gap

The literature identifies three dominant categories of retrieval approaches: keyword-based retrieval, semantic retrieval, and fuzzy retrieval.

Keyword-based retrieval approaches, including TF-IDF and BM25, remain widely used due to their efficiency and simplicity. However, they primarily rely on lexical matching and may struggle to address ambiguity, vocabulary mismatch, and exploratory search behavior [35, 34].

Recent advances in Natural Language Processing(NLP) have introduced semantic retrieval methods based on embedding models such as Sentence-BERT(SBERT). These approaches improve the ability to capture semantic relationships between queries and documents by representing text within continuous vector spaces [33]. While semantic retrieval reduces vocabulary mismatch, it often introduces additional complexity and reduced interpretability.

In contrast, fuzzy Information Retrieval(IR) provides mechanisms for modeling gradual relevance and uncertainty. Through fuzzy membership functions and rule-based reasoning, retrieval systems can represent relevance as a degree rather than a binary condi-

tion. Furthermore, fuzzy inference systems provide explicit reasoning structures that can be translated into understandable explanations [22, 42].

Although previous studies have investigated keyword-based retrieval, semantic retrieval, and fuzzy IR independently, the fundamental challenge remains insufficiently addressed. Existing approaches generally optimize lexical or semantic matching but provide limited support for combining multiple metadata-based relevance criteria within an interpretable decision-making process. As a result, users often receive ranked results without a transparent explanation of why particular datasets are prioritised, which is especially problematic in public-sector information systems where accountability and trust are important.

The research gap addressed in this thesis is therefore not merely the absence of a particular combination of techniques, but the lack of a human-centered retrieval framework that integrates multiple metadata-derived relevance signals while preserving transparency and explainability. This thesis addresses that gap through the design, implementation, and evaluation of a fuzzy logic-based HCIR framework for Swiss OGD, enabling interpretable multi-criteria ranking that supports exploratory dataset discovery and provides understandable explanations for ranking decisions.

1.4 Research Objectives

The primary objective of this thesis is to design, implement, and evaluate a fuzzy logic-based retrieval framework that improves dataset discoverability within the Swiss OGD portal.

The specific objectives are:

1. To analyze limitations of existing OGD retrieval approaches.
2. To design a metadata-driven retrieval framework capable of integrating multiple relevance criteria.
3. To develop a fuzzy inference model that supports gradual relevance estimation.
4. To generate interpretable explanations for ranking decisions.
5. To evaluate the effectiveness, usability, and transparency of the proposed system through quantitative and qualitative methods.
6. To investigate the contribution of fuzzy logic to HCIR.

Beyond defining the technical objectives of the proposed framework, it is equally important to identify the intended users and explain how their requirements influenced the overall system design. The proposed framework is primarily intended to support

researchers, public-sector analysts, journalists, students, and citizens seeking to discover and understand Swiss OGD. These user groups frequently perform exploratory searches in which information needs may be incomplete, ambiguous, or evolve during the search process. Consequently, the framework was designed according to HCIR principles, emphasizing transparent ranking decisions, explainable retrieval results, and the integration of multiple relevance criteria beyond keyword matching. These design choices aim to improve users' understanding of why datasets are retrieved and support more informed exploration of public-sector information.

1.5 Research Questions

The study is guided by the following research questions:

RQ1: How can fuzzy logic be used to improve dataset retrieval within OGD portals?

RQ2: How can multiple metadata-based relevance criteria be integrated into a unified and interpretable ranking mechanism?

RQ3: To what extent does a fuzzy logic-based retrieval system support HCIR principles?

RQ4: How effective are metadata-driven explanations in improving user understanding and trust in ranking decisions?

1.6 Research Contributions

This thesis makes several contributions to the fields of OGD retrieval, IR, and HCIR.

The first contribution is the development of a fuzzy logic-based retrieval framework specifically designed for OGD environments.

The second contribution is the integration of BM25-based similarity estimation with fuzzy logic-based multi-criteria relevance modeling using similarity, metadata completeness, dataset recency, and resource availability.

The third contribution is the introduction of an explainable ranking mechanism that generates human-readable explanations derived from dataset metadata.

The fourth contribution is a comprehensive evaluation framework that combines quantitative IR benchmarking with qualitative user-centered assessment. Retrieval effectiveness is evaluated by comparing the proposed fuzzy logic-based ranking framework with established baseline retrieval systems using standard IR metrics, while user-centered evaluation investigates usability, transparency, explainability, and user trust through structured participant studies. This mixed-method evaluation provides complementary evidence regarding both ranking performance and human-centered interaction.

Finally, the thesis contributes to HCIR research by demonstrating how fuzzy logic can support exploratory search, gradual relevance estimation, transparency, and user trust.

1.7 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 presents the literature review and theoretical background covering OGD, IR, HCIR, BM25 ranking, fuzzy logic, and explainable retrieval systems.

Chapter 3 describes the Design Science Research (DSR) methodology adopted in this study and provides a detailed justification for selecting DSR over alternative research methodologies.

Chapter 4 presents the architecture and design of the proposed retrieval framework, including metadata processing, feature engineering, fuzzy inference, and explainability mechanisms.

Chapter 5 discusses the implementation of the prototype and the technologies used to realize the proposed system.

Chapter 6 presents the complete evaluation methodology adopted for this research. The evaluation consists of two complementary components: a quantitative retrieval evaluation comparing the proposed framework against baseline retrieval approaches using standard IR metrics, and a qualitative user-centered evaluation investigating usability, transparency, explainability, and user trust through participant studies.

Chapter 7 presents and discusses the evaluation results. Quantitative retrieval performance is first analyzed using IR metrics, followed by the statistical analysis of questionnaire responses and qualitative participant feedback. The chapter concludes by discussing the implications of the findings with respect to the proposed research questions, HCIR principles, and the identified limitations of the study.

Finally, Chapter 8 summarizes the contributions of the thesis and outlines future directions for extending the proposed approach.

Chapter 2

Literature Review and Background

This chapter reviews the theoretical foundations and related research relevant to the proposed study. It examines existing work in OGD, IR, HCIR, fuzzy logic, and explainable retrieval systems. The chapter also analyzes the strengths and limitations of existing approaches and identifies the research gap that motivates the development of the proposed fuzzy logic-based retrieval framework.

2.1 Introduction

This chapter presents the theoretical foundations and related research relevant to the development of the proposed HCIR (HCIR) system for Swiss OGD. The purpose of this chapter is to establish the conceptual background of the study, review existing approaches to dataset retrieval, and identify the research gap addressed by this thesis.

The increasing availability of OGD has created significant opportunities for transparency, innovation, and evidence-based decision-making. However, the growing volume of datasets has also introduced challenges related to dataset discoverability and retrieval effectiveness. Users interacting with OGD portals often encounter difficulties when attempting to locate relevant datasets due to inconsistent metadata, vocabulary mismatch, and the exploratory nature of many search tasks [1, 26].

IR research has traditionally focused on keyword-based ranking techniques such as TF-IDF and BM25, which estimate relevance through statistical relationships between query terms and document content [35, 34]. Although these approaches have demonstrated strong performance in many retrieval scenarios, they often struggle to accommodate vague, incomplete, or evolving information needs.

Recent developments in NLP have introduced semantic retrieval methods based on vector embeddings and transformer architectures. Models such as SBERT enable retrieval systems to capture semantic similarity beyond exact keyword matching [33]. However, these approaches often operate as complex black-box systems, making it difficult to ex-

plain why particular results are ranked highly [9].

HCIR emphasizes the importance of designing retrieval systems that support human reasoning, exploratory search behavior, transparency, and trust [26]. Within this context, explainability becomes a critical design requirement, particularly in public-sector information systems where users may require understandable justifications for ranking decisions.

Fuzzy logic offers a promising framework for addressing these challenges. By representing relevance as a gradual rather than binary concept, fuzzy systems can model uncertainty, partial relevance, and subjective interpretations of information needs [42, 22]. Furthermore, fuzzy inference systems provide explicit rule-based reasoning mechanisms that can be translated into human-readable explanations, thereby supporting transparency and interpretability.

This chapter reviews the key concepts underlying OGD, IR, HCIR, fuzzy logic, and explainable retrieval systems. The chapter concludes by identifying limitations in existing approaches and establishing the research gap that motivates the proposed fuzzy logic-based retrieval framework.

2.2 Open Government Data

OGD has emerged as a key component of modern digital government initiatives. By providing public access to government-generated data, OGD promotes transparency, accountability, citizen participation, and innovation. Understanding the principles, benefits, and challenges of OGD is essential because the proposed retrieval system operates within the context of the Swiss national OGD portal, *opendata.swiss*. This section introduces the concept of OGD, discusses its objectives and benefits, and examines the challenges associated with managing and discovering government datasets.

2.2.1 Open Government Data: Concepts, Benefits, and Challenges

OGD refers to data produced, collected, or commissioned by public authorities that is made freely available for public access, use, reuse, and redistribution with minimal legal, technical, or financial restrictions [30, 31]. The objective of OGD initiatives is to increase transparency, encourage citizen participation, stimulate innovation, and improve the accessibility of public-sector information. By making government-generated data openly available, public institutions can support evidence-based decision-making while fostering greater engagement between governments, organizations, researchers, and citizens.

The concept of OGD is closely associated with the broader principles of open government, which emphasize transparency, accountability, and collaboration. According to

the Open Knowledge Foundation, data can be considered open when anyone can freely access, use, modify, and share it subject only to requirements such as attribution and preservation of openness [31]. Governments worldwide have adopted OGD strategies to improve public access to information and to encourage the development of innovative data-driven services and applications.

OGD provides several important benefits. First, it promotes transparency by allowing citizens to access information regarding governmental activities, expenditures, policies, and decision-making processes. Increased transparency can strengthen public trust and improve governmental accountability. Second, OGD supports innovation and economic development by enabling researchers, entrepreneurs, and organizations to create new products, services, and analytical solutions using publicly available data [18]. Third, OGD contributes to evidence-based policymaking by providing access to datasets that support research, policy evaluation, and informed decision-making across multiple domains [18].

Despite these advantages, several challenges continue to limit the effective use of OGD. One of the most significant challenges is dataset discoverability. As the number of published datasets increases, users often struggle to locate relevant information within large repositories containing thousands of datasets [21]. Traditional keyword-based retrieval systems may not adequately support exploratory search activities or complex information needs, particularly when users are unfamiliar with available datasets.

Another challenge concerns metadata quality and consistency. Since datasets are published by different governmental organizations, metadata descriptions often vary in completeness, structure, and terminology. Some datasets contain detailed descriptions, keywords, and documentation, whereas others provide only limited metadata. These inconsistencies can negatively affect retrieval effectiveness and make it difficult for users to evaluate dataset relevance.

Data heterogeneity presents an additional challenge for OGD portals. OGD repositories typically contain datasets from diverse domains such as transportation, health, environment, energy, education, economics, and public administration. These datasets differ substantially in content, format, vocabulary, and update frequency, increasing the complexity of retrieval and ranking processes.

Furthermore, users frequently approach OGD portals with vague, evolving, or exploratory information needs. Rather than searching for a specific known dataset, users often seek to discover information, compare alternatives, or explore unfamiliar topics. Such search behavior introduces uncertainty into the retrieval process and highlights the limitations of retrieval approaches that rely solely on exact keyword matching.

These challenges demonstrate that effective OGD retrieval requires more than conventional search functionality. Retrieval systems must support dataset discoverability, accommodate metadata variability, provide meaningful relevance estimation, and assist users in understanding ranking decisions. These requirements motivate the exploration of

HCIR and fuzzy logic-based approaches, which aim to improve both retrieval effectiveness and user understanding within OGD environments.

2.2.2 Open Government Data in Switzerland

Switzerland supports OGD through the national portal *opendata.swiss*, which serves as a centralized platform for accessing datasets published by federal, cantonal, and municipal institutions [11]. The portal provides access to thousands of datasets covering diverse domains such as transportation, environment, energy, health, population, economy, and public administration.

The portal is built on the CKAN platform and follows the DCAT-AP CH metadata standard, enabling standardized dataset representation and interoperability across organizations. Users can access datasets through both a web interface and APIs, facilitating automated data retrieval and integration.

The large volume of datasets and the diversity of metadata descriptions available on *opendata.swiss* make it an appropriate environment for studying dataset discoverability and evaluating advanced retrieval approaches. Consequently, the portal serves as the primary data source for the retrieval framework developed in this thesis.

2.3 Dataset Discoverability in Open Government Data Portals

The success of OGD initiatives depends not only on the publication of datasets but also on users' ability to discover and access relevant information efficiently. As OGD repositories continue to grow in size and diversity, dataset discoverability has become a major challenge for governments, researchers, and information system designers. Effective discoverability enables users to locate relevant datasets quickly, understand their content, and assess their suitability for specific information needs.

Dataset discoverability is closely linked to metadata quality, retrieval effectiveness, and user interaction. Since users often rely on metadata rather than dataset contents during search, retrieval systems must support efficient metadata exploration while accommodating diverse search behaviors and varying levels of user expertise.

2.3.1 Metadata-Based Retrieval and Discoverability Challenges

Most OGD portals rely on metadata-based retrieval mechanisms [21]. Instead of searching the contents of datasets directly, retrieval systems typically index metadata attributes such as titles, descriptions, keywords, categories, publishers, and update information.

Metadata therefore serves as the primary source of information used to estimate dataset relevance and support user search activities.

Metadata-based retrieval offers several advantages. It enables efficient indexing of large repositories, reduces computational complexity, and provides descriptive information that helps users evaluate datasets before downloading or analyzing them. Consequently, metadata plays a critical role in supporting dataset discovery and information access within OGD environments [28].

However, the effectiveness of metadata-based retrieval depends heavily on metadata quality. In many OGD repositories, metadata is created by multiple organizations using different standards, vocabularies, and levels of detail. As a result, metadata records often vary considerably in completeness, consistency, and descriptive quality.

Incomplete metadata represents a particularly significant challenge. Datasets lacking meaningful descriptions, keywords, or update information may be difficult to retrieve even when they are highly relevant to user queries. Similarly, inconsistent terminology can reduce retrieval effectiveness by preventing accurate matching between user information needs and dataset descriptions.

Another challenge arises from the scale and heterogeneity of OGD repositories. Modern portals frequently contain thousands of datasets spanning diverse domains such as transportation, health, environment, energy, demographics, and public administration. The diversity of topics and metadata structures increases the complexity of retrieval and ranking processes.

These challenges highlight the need for retrieval approaches that extend beyond traditional keyword matching and incorporate additional indicators of relevance, quality, and usefulness.

2.3.2 Exploratory Search in Open Government Data Environments

Unlike traditional lookup searches, exploratory search occurs when users seek to learn, investigate, compare, or discover information rather than retrieve a specific known item [26]. Exploratory search is particularly common within OGD environments because users are often unfamiliar with available datasets and may not know precisely what information exists within the repository.

In many cases, users begin with broad or vague information needs that evolve throughout the search process. They may refine queries, examine multiple datasets, compare metadata descriptions, and gradually develop a better understanding of available information resources [26]. This iterative behavior differs substantially from conventional keyword-based retrieval scenarios.

Supporting exploratory search requires retrieval systems that accommodate uncer-

tainty and provide meaningful guidance during information seeking. Users benefit from ranking mechanisms that consider multiple relevance factors rather than relying exclusively on lexical similarity. Furthermore, transparency and explainability become increasingly important because users often need to understand why datasets were recommended and how ranking decisions were made.

These requirements align closely with the principles of HCIR, which emphasize user understanding, interaction, transparency, and support for complex information-seeking behavior. Consequently, exploratory search provides a strong motivation for developing retrieval frameworks that combine effective ranking mechanisms with explainable and user-centered design principles.

2.4 Information Retrieval Models

IR systems are designed to identify and rank documents according to their relevance to user information needs. Within OGD environments, retrieval systems play an important role in helping users locate datasets from large and heterogeneous repositories. Effective retrieval models must therefore support accurate relevance estimation while remaining computationally efficient and interpretable.

Over the years, numerous retrieval approaches have been proposed, ranging from traditional lexical matching methods to modern semantic retrieval techniques based on deep learning and language models. Understanding the strengths and limitations of these approaches is important for selecting an appropriate retrieval foundation for HCIR systems.

2.4.1 Traditional Information Retrieval Models

Traditional IR approaches estimate relevance primarily through lexical matching between query terms and document content. These methods assume that documents containing terms similar to those used in a query are more likely to satisfy user information needs.

One of the earliest and most influential approaches is Term Frequency–Inverse Document Frequency (TF-IDF), which assigns weights to terms based on their frequency within a document and their rarity across the collection [35]. Although TF-IDF remains widely used, it does not explicitly account for document length normalization or term frequency saturation.

To address these limitations, probabilistic retrieval models such as BM25 were developed. BM25 extends traditional term-weighting approaches by incorporating inverse document frequency, term frequency saturation, and document length normalization. As a result, BM25 has become one of the most widely adopted retrieval models in both academic research and real-world search systems [34].

The BM25 ranking function is defined as:

$$BM25(D, Q) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, D)(k_1 + 1)}{f(q_i, D) + k_1 \left(1 - b + b \frac{|D|}{avgdl}\right)} \quad (2.1)$$

where $f(q_i, D)$ represents the frequency of query term q_i in document D , $|D|$ denotes document length, $avgdl$ is the average document length within the collection, and k_1 and b are model parameters controlling term frequency saturation and document length normalization.

Due to its effectiveness, simplicity, and interpretability, BM25 continues to serve as a strong baseline for many retrieval applications.

2.4.2 Semantic Retrieval Models

Recent advances in NLP have led to the development of semantic retrieval approaches that represent documents and queries using dense vector embeddings. Unlike traditional lexical methods, semantic retrieval models attempt to capture contextual meaning and conceptual relationships between terms.

Transformer-based models such as SBERT generate dense vector representations that can be compared using similarity measures to identify semantically related documents [33]. These approaches can improve retrieval performance when relevant documents use different terminology than the submitted query.

Despite their advantages, semantic retrieval models present several challenges. They typically require substantial computational resources, large pre-trained models, and additional infrastructure for vector indexing and similarity search. Furthermore, the reasoning process underlying retrieval decisions is often less transparent than traditional lexical approaches, making explainability more difficult.

For applications emphasizing transparency, interpretability, and explainable decision-making, traditional retrieval models remain attractive alternatives.

2.4.3 Selection of BM25 for the Proposed Framework

Several retrieval approaches were considered during the design of the proposed framework, including TF-IDF, cosine similarity-based retrieval, dense embeddings, and transformer-based semantic retrieval models. However, BM25 was ultimately selected as the primary retrieval mechanism due to its effectiveness, interpretability, and compatibility with HCIR objectives.

Compared with TF-IDF, BM25 provides more sophisticated relevance estimation through document length normalization and term frequency saturation. These mechanisms reduce biases toward longer documents and generally produce more robust ranking

performance [34].

Dense embedding approaches and transformer-based models such as SBERT offer strong semantic capabilities but introduce additional computational complexity and reduced interpretability. Since one of the primary objectives of this research is to support transparency and explainability, highly complex black-box retrieval models were considered less suitable for the proposed framework.

Another important consideration was integration with fuzzy logic-based ranking. BM25 produces interpretable relevance scores that can be incorporated directly into a fuzzy inference system and explained to users. This compatibility supports HCIR principles by enabling transparent relevance estimation and understandable ranking decisions.

Cosine similarity was also considered as a potential retrieval approach. However, cosine similarity primarily measures the angular similarity between vector representations and does not inherently account for document length normalization, term frequency saturation, or probabilistic relevance estimation. In contrast, BM25 was specifically designed for IR applications and has consistently demonstrated strong performance across benchmark retrieval tasks. Furthermore, BM25 provides interpretable relevance scores that can be integrated directly into the fuzzy inference system developed in this research. For these reasons, BM25 was considered more suitable than cosine similarity for metadata-based dataset retrieval within OGD environments [34].

Although recent neural retrieval methods such as Dense Passage Retrieval (DPR), ColBERT, and Large Language Model (LLM)-based rerankers have demonstrated strong retrieval performance, they generally require substantially larger training corpora, greater computational resources, and often provide limited transparency regarding ranking decisions [19, 20, 23, 37]. In contrast, the objective of this research is not solely to maximise retrieval effectiveness but to investigate how multiple metadata-derived relevance criteria can be integrated within an interpretable HCIR framework. BM25 therefore provides an appropriate lexical retrieval foundation because it is computationally efficient, reproducible, and widely adopted as a strong baseline in IR research. Its transparent scoring mechanism also allows the subsequent fuzzy inference layer to explain how different metadata characteristics contribute to the final ranking, which would be considerably more difficult with many neural retrieval approaches.

2.5 HCIR

Traditional IR research has primarily focused on improving ranking effectiveness and retrieval accuracy. While these objectives remain important, modern retrieval systems must also consider how users interact with information systems, interpret retrieval outcomes, and make decisions based on retrieved information.

HCIR extends traditional retrieval approaches by placing users at the center of the

retrieval process. Rather than treating retrieval as a purely algorithmic task, HCIR emphasizes interaction, transparency, interpretability, and support for human decision-making. The objective is not only to retrieve relevant information but also to help users understand, explore, and effectively utilize retrieval results.

2.5.1 Principles of HCIR

HCIR is founded on the recognition that information seeking is an interactive and evolving process [36, 26]. Users frequently refine queries, explore alternative information sources, and modify search objectives as their understanding develops. Consequently, retrieval systems should support exploration and learning rather than merely returning ranked lists of documents.

Several principles characterize HCIR. First, retrieval systems should support user interaction by enabling exploration, query refinement, and iterative search behavior. Second, systems should provide meaningful feedback that helps users understand search results and navigate information spaces effectively. Third, retrieval mechanisms should accommodate uncertainty because users often begin searches with incomplete or evolving information needs.

Unlike purely system-centered approaches, HCIR recognizes that successful retrieval depends not only on algorithmic performance but also on user understanding, satisfaction, and trust. These principles are particularly relevant within OGD environments where users frequently engage in exploratory search activities and may possess varying levels of technical expertise.

2.5.2 Transparency and Explainability in HCIR

Transparency and explainability represent fundamental objectives of HCIR [36]. Users are more likely to trust retrieval systems when they can understand how ranking decisions are generated and why particular results are recommended.

Traditional retrieval systems often provide limited insight into the reasoning process underlying ranking outcomes. As a result, users may find it difficult to evaluate the reliability or relevance of retrieved information. Explainable retrieval systems address this challenge by providing understandable descriptions of ranking decisions and relevance estimation processes.

Within HCIR, explainability supports several important objectives. First, it improves user understanding by making retrieval processes more accessible. Second, it enhances trust by allowing users to assess the reasoning behind recommendations. Third, it supports learning and exploration by helping users understand relationships between queries, metadata attributes, and retrieval outcomes.

Consequently, transparency and explainability are increasingly recognized as essential characteristics of user-centered retrieval systems.

2.5.3 HCIR in Open Government Data Environments

The principles of HCIR are particularly relevant within OGD environments [21]. Users of OGD portals often possess diverse backgrounds, varying levels of technical expertise, and different information needs. Furthermore, many users engage in exploratory search activities and may be unfamiliar with available datasets or domain-specific terminology.

Traditional keyword-based retrieval systems may struggle to support these complex information-seeking behaviors because they focus primarily on lexical matching rather than user understanding. HCIR approaches seek to address these limitations by emphasizing interaction, transparency, explainability, and support for exploratory search.

In the context of this research, HCIR provides the conceptual foundation for the development of an explainable retrieval framework that combines BM25 retrieval, fuzzy logic-based ranking, and metadata-driven explanations. By integrating these components, the proposed framework aims to improve both retrieval effectiveness and user understanding within the Swiss OGD portal.

2.6 Fuzzy Logic and Information Retrieval

Traditional retrieval systems frequently represent relevance as a binary concept, where documents are considered either relevant or non-relevant. However, human judgments rarely operate in such a strictly categorical manner. Users often perceive relevance as gradual, uncertain, and context-dependent. Consequently, alternative approaches are required to model the ambiguity and subjectivity inherent in information-seeking behavior.

Fuzzy logic provides a mathematical framework for representing uncertainty and gradual reasoning. Introduced by Zadeh [42], fuzzy logic allows concepts to be represented through varying degrees of membership rather than binary classifications. This capability makes fuzzy logic particularly attractive for IR applications where relevance is often subjective and difficult to define precisely.

2.6.1 Fundamentals of Fuzzy Logic

Unlike classical Boolean logic, which assigns elements exclusively to either membership or non-membership categories, fuzzy logic allows partial membership. Membership values typically range between 0 and 1, representing the degree to which an element belongs to a particular fuzzy set [42].

Fuzzy logic utilizes linguistic variables such as Low, Medium, and High to describe concepts that cannot be defined precisely. Membership functions are used to map nu-

merical values into these linguistic categories, thereby enabling gradual reasoning and the representation of uncertainty [42].

This flexibility allows fuzzy systems to model complex human judgments more naturally than traditional binary approaches. Consequently, fuzzy logic has been widely applied in decision support systems, expert systems, control systems, and IR applications.

2.6.2 Fuzzy Inference Systems

A fuzzy inference system combines fuzzy sets, linguistic variables, and rule-based reasoning to generate decisions or relevance estimates. The process typically consists of four stages: fuzzification, rule evaluation, aggregation, and defuzzification.

Among the available inference approaches, the Mamdani model is one of the most widely used because of its interpretability and compatibility with human reasoning [24]. Mamdani systems employ IF–THEN rules that describe relationships between input variables and output decisions using linguistic terms.

For example:

IF Similarity is High AND Metadata Completeness is High THEN Relevance is Very High.

Such rules resemble human decision-making processes and therefore contribute to transparency and explainability.

2.6.3 Fuzzy Logic in Information Retrieval

Fuzzy logic has been applied extensively within IR research to address limitations associated with binary relevance judgments [4]. Rather than classifying documents as simply relevant or non-relevant, fuzzy retrieval approaches represent relevance as a gradual concept characterized by varying degrees of membership.

Several studies have demonstrated that fuzzy logic can improve retrieval performance by integrating multiple relevance criteria and accommodating uncertainty within user information needs. Fuzzy approaches are particularly useful in environments where relevance depends on multiple factors rather than a single similarity measure.

Within OGD repositories, dataset relevance may depend not only on textual similarity but also on metadata quality, recency, completeness, and contextual suitability. Fuzzy inference systems provide a natural mechanism for combining these heterogeneous relevance indicators into a unified ranking framework.

2.6.4 Why Fuzzy Logic Supports HCIR

One of the central arguments of this thesis is that fuzzy logic provides characteristics that align closely with HCIR principles.

First, fuzzy logic models gradual relevance rather than binary relevance [4]. Human users rarely perceive information as completely relevant or completely irrelevant. Instead, relevance is often evaluated along a continuum influenced by context, task requirements, and personal preferences. Fuzzy logic naturally supports this type of reasoning.

Second, fuzzy inference systems accommodate uncertainty and ambiguity [42]. Users frequently submit incomplete, vague, or exploratory queries, particularly within OGD environments. Fuzzy reasoning enables retrieval systems to handle such uncertainty more effectively than rigid rule-based approaches.

Third, fuzzy logic promotes transparency and explainability through explicit IF-THEN rules [36]. Unlike many black-box machine learning models, fuzzy systems allow users to understand how relevance estimates are generated and which factors contribute to ranking decisions.

Finally, fuzzy logic facilitates the integration of multiple relevance indicators. In the proposed framework, thematic similarity, metadata completeness, and dataset recency are combined through fuzzy reasoning to generate relevance estimates that more closely reflect human judgment.

For these reasons, fuzzy logic represents a suitable foundation for HCIR systems that emphasize transparency, interpretability, and user understanding.

From a HCIR perspective, fuzzy logic enables retrieval systems to operate in a manner that more closely resembles human reasoning. Human users often evaluate relevance using imprecise concepts such as "mostly relevant", "somewhat useful", or "highly related" rather than strict numerical thresholds. Fuzzy logic allows these gradual judgments to be represented mathematically through linguistic variables and membership functions. Consequently, fuzzy reasoning provides a natural bridge between computational ranking mechanisms and human relevance perception, making it particularly suitable for HCIR environments [3, 42].

While alternative approaches such as weighted-sum ranking, learning-to-rank models, and multi-criteria decision-making methods (e.g., AHP or TOPSIS) can also combine multiple relevance factors, they present different trade-offs. Weighted-sum approaches assume fixed linear relationships between ranking criteria and cannot naturally represent gradual transitions between linguistic concepts such as *low*, *medium*, and *high* relevance. Learning-to-rank methods generally require substantially larger collections of labelled training data than were available in this study and often reduce transparency by producing complex learned ranking functions. Similarly, multi-criteria decision-making methods are well suited for structured decision problems but typically provide less flexibility for

modelling uncertainty and subjective relevance judgments. Fuzzy inference was therefore selected because it supports interpretable rule-based reasoning while allowing multiple metadata-derived relevance criteria to be combined in a manner that closely reflects human reasoning under uncertainty.

Among the available fuzzy inference approaches, the Mamdani model was selected in preference to Sugeno or Takagi–Sugeno–Kang (TSK) inference because the primary objective of this research is interpretability rather than computational optimisation. Mamdani inference produces linguistic output membership functions that can be directly translated into human-readable explanations, thereby supporting transparency and explainability within HCIR. Although Sugeno-based systems are computationally efficient and are often preferred for optimisation and control applications, their numerical outputs are generally less intuitive for explaining ranking decisions to end users. Consequently, Mamdani inference provides a more appropriate foundation for the proposed explainable retrieval framework [38, 42].

2.7 Explainable Information Retrieval

As retrieval systems become increasingly complex, users often find it difficult to understand why particular documents or datasets are recommended. Although advanced ranking algorithms may improve retrieval effectiveness, a lack of transparency can reduce user trust and hinder effective decision-making. Consequently, explainability has emerged as an important research area within IR.

Explainable Information Retrieval (XIR) seeks to provide users with understandable descriptions of ranking decisions and retrieval outcomes. Rather than presenting search results as opaque recommendations, explainable systems communicate the factors that influenced relevance estimation and document ranking. This transparency helps users interpret retrieval results and assess their suitability for specific information needs.

2.7.1 Explainability in Information Retrieval

Explainability refers to the ability of a system to communicate the reasoning process underlying its decisions in a manner that is understandable to human users [27]. Within IR, explainability aims to clarify why particular documents are retrieved, how relevance scores are generated, and which factors contribute to ranking decisions.

Explainable retrieval systems can provide information regarding query-document similarity, metadata characteristics, relevance indicators, and ranking criteria. Such explanations help users understand retrieval outcomes and improve confidence in system recommendations.

2.7.2 Benefits and Challenges of Explainability

Explainability provides several important benefits within retrieval systems [27]. First, it improves transparency by exposing the factors that influence ranking decisions. Second, it supports user trust by enabling users to understand and evaluate retrieval outcomes. Third, explanations can facilitate learning and exploratory search by helping users understand relationships between queries and retrieved information.

Despite these advantages, implementing explainability presents several challenges. Explanations must remain accurate while also being understandable to users with varying levels of expertise. Excessively technical explanations may confuse users, whereas overly simplified explanations may fail to communicate meaningful information.

Designing effective explanations therefore requires balancing transparency, usability, and informational value. HCIR emphasizes this balance by focusing on explanations that support user understanding and decision-making.

2.7.3 Explainability for Open Government Data Retrieval

Explainability is particularly valuable within OGD environments because users often possess limited knowledge regarding available datasets, metadata structures, and retrieval mechanisms. Consequently, users may find it difficult to determine why specific datasets are recommended or how relevance scores are generated.

Providing explanations alongside retrieval results can improve user understanding of dataset characteristics and assist users in evaluating dataset suitability [27]. Metadata-driven explanations are especially useful because they communicate observable factors such as thematic similarity, metadata completeness, and dataset recency.

In the proposed framework, explainability is achieved through the integration of fuzzy logic-based reasoning and metadata-driven relevance estimation. The use of explicit fuzzy rules enables ranking decisions to be communicated in a transparent and understandable manner, thereby supporting HCIR objectives and improving user trust in retrieval outcomes.

Unlike post-hoc explanation methods that generate explanations independently of the underlying ranking process, the proposed framework produces *faithful* explanations that directly reflect the reasoning performed by the fuzzy inference system. The generated explanations are derived from the same metadata features, fuzzy membership functions, and rule activations used to compute the final relevance score. Consequently, the explanation presented to the user corresponds directly to the actual ranking process rather than providing a separate or approximate interpretation. This close correspondence between ranking and explanation supports transparency, accountability, and user trust, which are fundamental objectives of HCIR and Explainable IR.

2.8 Research Gap and Conceptual Framework

The literature reviewed in this chapter demonstrates significant progress in OGD retrieval, IR, HCIR, fuzzy logic, and explainable retrieval systems. However, several limitations remain that motivate the development of the proposed framework.

Although existing studies provide valuable contributions, relatively few approaches integrate retrieval effectiveness, explainability, transparency, and user-centered design within a unified OGD retrieval framework. Furthermore, limited research has explored the combined use of BM25 retrieval, fuzzy logic-based relevance estimation, and HCIR principles for dataset discovery. These observations form the basis for the research gap addressed by this thesis.

2.8.1 Research Gap

The literature reveals several limitations in current OGD retrieval systems.

First, many OGD portals rely primarily on traditional keyword-based retrieval mechanisms. While such approaches can efficiently identify datasets containing matching query terms, they often struggle to accommodate exploratory search behavior, uncertainty, and evolving information needs. Consequently, retrieval effectiveness may be limited when users are unfamiliar with available datasets or employ vocabulary different from that used in metadata descriptions.

Second, existing retrieval approaches frequently focus on textual similarity while overlooking additional factors that may influence dataset relevance. Characteristics such as metadata completeness, dataset recency, and metadata quality can affect the usefulness of datasets but are often excluded from ranking processes.

Third, explainability remains limited within many retrieval environments. Users are typically presented with ranked results without receiving meaningful explanations regarding how relevance scores were generated. This lack of transparency can reduce trust and make it difficult for users to evaluate retrieval outcomes.

Finally, despite increasing interest in HCIR, relatively little research has investigated the integration of HCIR principles, fuzzy logic-based reasoning, and OGD retrieval within a unified framework. Existing studies tend to focus on individual aspects of retrieval rather than providing a comprehensive solution that combines retrieval effectiveness, transparency, explainability, and user-centered interaction.

These limitations highlight the need for retrieval frameworks capable of supporting both effective dataset discovery and meaningful user understanding.

More specifically, no identified study was found that combines BM25 retrieval, metadata completeness assessment, dataset recency evaluation, fuzzy logic-based relevance estimation, and explainable HCIR within the context of the Swiss OGD portal [34, 4, 36].

This specific combination constitutes the primary research gap addressed by the present study.

2.8.2 Conceptual Framework

To address the identified research gap, this thesis proposes a HCIR framework for Swiss OGD. The framework integrates BM25-based retrieval, fuzzy logic-based relevance estimation, metadata-driven ranking, explainability mechanisms, and a user-centered interface within a unified retrieval architecture [34, 42, 3].

The framework is designed to improve dataset discoverability while supporting transparency, explainability, and user understanding. Instead of relying solely on textual similarity, the framework combines multiple relevance indicators, including thematic similarity, metadata completeness, dataset recency, and resource availability to produce relevance estimates that more closely reflect human judgment.

Table 2.1 summarizes the major components of the proposed framework and their respective roles.

Table 2.1: Conceptual Framework Components

Component	Purpose
Metadata Acquisition and Processing	Collects and preprocesses metadata from the Swiss OGD portal.
BM25 Retrieval Engine	Calculates thematic similarity between user queries and dataset metadata.
Fuzzy Inference System	Combines thematic similarity, metadata completeness, and dataset recency and resource availability to estimate relevance.
Explainability Module	Generates understandable explanations for ranking decisions.
User Interface(UI)	Supports user interaction, dataset exploration, and result visualization.

The interaction among these components enables the framework to move beyond traditional keyword-based retrieval approaches. By integrating retrieval effectiveness with explainability and user-centered design principles, the proposed framework aims to improve dataset discovery and support more effective interaction with OGD resources.

2.9 Chapter Summary

This chapter reviewed the literature related to OGD, dataset discoverability, IR, HCIR, fuzzy logic, and explainable retrieval systems. The review identified limitations in existing OGD retrieval approaches, particularly regarding transparency, explainability, and support for exploratory search. Based on these observations, a HCIR framework integrating BM25 retrieval, fuzzy logic-based ranking, and explainability mechanisms was proposed to address the identified research gap.

Chapter 3

Research Methodology

This chapter describes the research methodology adopted to guide the design, development, and evaluation of the proposed retrieval framework. The chapter introduces the DSR paradigm, explains its suitability for the objectives of this study, and presents the research process followed throughout the project. In addition, the chapter discusses alternative research methodologies and justifies the selection of DSR as the most appropriate methodological approach for developing and evaluating a HCIR system for Swiss OGD.

3.1 Introduction to Research Methodology

Research methodology provides the systematic framework through which scientific investigations are conducted. The choice of an appropriate methodology is critical because it influences how research questions are addressed, how data is collected and analyzed, and how research contributions are evaluated. Different methodologies are suitable for different research objectives, ranging from theory development and hypothesis testing to artifact design and practical problem solving.

The objective of this thesis is not only to study an existing phenomenon but also to design, implement, and evaluate a novel retrieval framework that addresses identified limitations in OGD retrieval systems. Consequently, a methodology capable of supporting artifact creation, iterative refinement, and practical evaluation is required. This requirement motivated the adoption of DSR as the primary methodological framework for the study.

3.2 Design Science Research

DSR is a research paradigm that focuses on the creation and evaluation of innovative artifacts designed to solve identified problems [14]. Unlike purely descriptive research approaches, DSR aims to generate practical solutions while simultaneously contributing

to scientific knowledge.

In Information Systems research, artifacts may take various forms, including models, frameworks, methods, algorithms, software systems, and design principles. The central objective of DSR is to develop artifacts that address real-world problems while providing insights that extend beyond the specific implementation.

According to Hevner et al. [14], DSR consists of iterative cycles involving problem identification, artifact design, implementation, evaluation, and refinement. Through these cycles, researchers continuously improve artifact quality while generating new knowledge regarding its effectiveness and applicability.

The proposed retrieval framework developed in this thesis constitutes a design artifact because it introduces a new combination of BM25 retrieval, fuzzy inference-based ranking, and explainable relevance estimation for OGD retrieval. Therefore, DSR provides an appropriate methodological foundation for guiding the development and evaluation process.

3.3 Why Design Science Research?

The selection of DSR was motivated by the objectives and nature of this study. The primary goal of the research is to design and evaluate a novel retrieval framework rather than merely observe or explain an existing phenomenon. Consequently, a methodology focused on artifact creation and evaluation is required.

First, DSR supports the development of innovative solutions to practical problems. The research addresses the challenge of dataset discoverability within OGD portals through the design of a fuzzy logic-based HCIR framework. Since the study seeks to produce a functional retrieval system, DSR aligns naturally with the research objectives.

Second, DSR encourages iterative development and refinement. The proposed retrieval framework underwent multiple stages of design, implementation, evaluation, and improvement throughout the project. This iterative process is consistent with the principles of DSR and allows continuous enhancement of artifact quality.

Third, DSR emphasizes both practical relevance and scientific rigor. The research contributes practically by improving dataset retrieval and contributes theoretically by demonstrating how fuzzy logic can support HCIR principles within OGD environments.

Finally, DSR is widely accepted within Information Systems research and has been successfully applied in studies involving decision support systems, recommendation systems, retrieval systems, and HCI. Therefore, DSR provides a well-established methodological framework for the development of the proposed retrieval artifact [14, 32].

3.3.1 Justification for Selecting Design Science Research

Several research methodologies were considered during the planning of this study, including experimental research, survey research, case study research, and DSR. Design Science Research was ultimately selected because the primary objective of the research was not only to study an existing phenomenon but also to design, develop, and evaluate a novel artifact capable of addressing an identified problem.

The central problem addressed by this research concerns dataset discoverability within the Swiss OGD portal. Existing retrieval approaches often provide limited support for exploratory search, explainability, and user-centered interaction. To address these limitations, this research proposes and implements a HCIR framework that combines BM25 retrieval, fuzzy logic-based relevance estimation, and explainability mechanisms.

Unlike purely observational methodologies, DSR explicitly focuses on the creation and evaluation of innovative artifacts that solve real-world problems [32]. The retrieval framework developed in this study constitutes such an artifact. Consequently, DSR provides a structured methodology for guiding artifact design, implementation, demonstration, and evaluation [32].

Alternative methodologies were considered less suitable for the objectives of this research. Experimental research primarily focuses on testing hypotheses rather than designing artifacts. Survey research is appropriate for collecting perceptions and opinions but does not provide guidance for system development. Case study research enables in-depth investigation of existing environments but does not emphasize artifact construction. In contrast, DSR directly supports the iterative design and evaluation of information systems intended to address practical challenges.

Therefore, DSR was selected because it provides a rigorous and widely accepted framework for developing and evaluating innovative information system artifacts while simultaneously contributing to scientific knowledge and practical problem solving.

3.4 Why Not Other Research Methodologies?

Several alternative research methodologies were considered before selecting DSR. Although these methodologies offer valuable perspectives, they were not fully aligned with the objectives of the present study.

3.4.1 Survey Research

Survey research focuses primarily on collecting opinions, attitudes, and perceptions from participants through questionnaires and interviews. While surveys are useful for understanding user behavior and preferences, they do not provide a structured framework for designing and evaluating technological artifacts.

Since the primary objective of this research is to develop and evaluate a retrieval system rather than solely investigate user perceptions, survey research alone would be insufficient. Nevertheless, survey techniques were employed during the evaluation phase to assess user satisfaction and usability.

3.4.2 Experimental Research

Experimental research is commonly used to test hypotheses by manipulating variables under controlled conditions. Although experimental methods can evaluate system performance, they do not provide guidance for artifact design and development.

The present study involves the creation of a retrieval framework rather than the testing of isolated causal relationships. Consequently, experimental research was incorporated as part of the evaluation process but was not selected as the overall methodological framework.

3.4.3 Case Study Research

Case study research aims to investigate phenomena within specific real-world contexts. While case studies provide rich contextual insights, they generally focus on understanding existing situations rather than creating new technological solutions.

The goal of this research is to develop and evaluate a novel retrieval framework rather than analyze a particular organizational setting. Therefore, case study research was not considered the most appropriate primary methodology.

3.4.4 Action Research

Action Research involves close collaboration between researchers and practitioners to solve organizational problems through cycles of intervention and reflection. Although Action Research supports practical problem solving, it typically requires continuous organizational participation throughout the research process.

The proposed retrieval framework was developed independently and evaluated through controlled user studies rather than organizational interventions. Consequently, DSR provided a more suitable methodological approach.

3.5 Design Science Research Process

The research process followed the DSR framework proposed by Hevner et al. [14]. The process consisted of five major phases:

1. Problem Identification

2. Objectives Definition
3. Artifact Design
4. Artifact Development
5. Evaluation and Refinement

The research process began with an analysis of dataset discoverability challenges within OGD portals. Following the identification of research gaps, research objectives were formulated to guide the design of a HCIR framework.

The retrieval artifact was subsequently designed and implemented using a custom BM25 retrieval component, a Mamdani fuzzy inference engine, and metadata-driven explanation mechanisms. The completed prototype was evaluated using a mixed-method evaluation strategy consisting of quantitative retrieval benchmarking and qualitative user-centered assessment. Quantitative evaluation compared the proposed framework with baseline retrieval approaches using standard IR metrics, while the qualitative evaluation examined usability, transparency, explainability, and user trust through structured participant studies. The combined findings were then used to assess the proposed framework and identify opportunities for further refinement.

3.5.1 Research Process Overview

The overall research process adopted in this study consisted of the following stages:

1. Literature Review
2. Problem Identification
3. Research Gap Analysis
4. Artifact Design
5. Prototype Development
6. System Evaluation
7. Refinement and Improvement
8. Final Analysis and Conclusions

The iterative nature of this process ensured that evaluation results continuously informed artifact improvements, thereby increasing both system quality and research rigor.

3.6 Chapter Summary

This chapter presented the research methodology adopted for the study and justified the selection of DSR as the guiding methodological framework. DSR was chosen because it supports the design, implementation, evaluation, and refinement of technological artifacts while maintaining scientific rigor and practical relevance.

The chapter also discussed alternative methodologies and explained why they were less suitable for achieving the objectives of the study. Finally, the DSR process followed throughout the project was presented. The next chapter describes the architecture and design of the proposed fuzzy logic-based HCIR framework.

Chapter 4

System Design

This chapter presents the architecture and design of the proposed HCIR framework for Swiss OGD. The chapter describes the overall system architecture, data flow, retrieval components, fuzzy inference framework, and explainability mechanisms. Particular attention is given to the integration of BM25 retrieval and fuzzy logic-based ranking, which together form the core of the proposed retrieval system.

4.1 Introduction

The design of an effective retrieval system requires the integration of multiple components that collectively support dataset discovery, relevance estimation, and user interaction. Based on the limitations identified in existing OGD retrieval systems, the proposed framework was designed to support exploratory search, transparent ranking, and explainable decision-making.

The system combines traditional IR techniques with fuzzy logic-based reasoning in order to provide a more human-centered retrieval experience. Rather than relying solely on keyword matching, the framework incorporates multiple metadata-driven relevance factors and produces interpretable explanations for ranking decisions. This section presents the overall design principles and architectural considerations that guided the development of the system.

4.2 Design Objectives

The design of the proposed retrieval framework was guided by several objectives derived from the research questions and literature review.

First, the system should improve dataset discoverability within OGD environments. Users should be able to locate relevant datasets efficiently even when queries are vague or exploratory.

Second, the framework should support HCIR principles by providing transparency, explainability, and user-friendly interaction mechanisms.

Third, the system should integrate multiple relevance criteria rather than relying solely on keyword similarity. Metadata completeness, dataset recency, and retrieval relevance should all contribute to ranking decisions.

Fourth, the framework should remain computationally efficient and scalable for practical deployment within OGD portals.

Finally, the retrieval process should be explainable so that users can understand why datasets are ranked in particular positions.

4.3 Overall System Architecture

The proposed system follows a layered architecture consisting of metadata acquisition, metadata processing, retrieval, fuzzy ranking, explainability, and user interaction components. User queries are processed through the retrieval layer, which estimates thematic similarity using BM25. Additional metadata quality indicators are then evaluated through a fuzzy inference system to generate final relevance scores. The explainability layer produces human-readable explanations that are presented through the UI. The proposed retrieval framework consists of five major components:

1. Dataset Acquisition Layer
2. Metadata Processing Layer
3. BM25 Retrieval Layer
4. Fuzzy Ranking Layer
5. Explainability and UI Layer

The retrieval process begins when a user submits a search query through the interface. The query is processed and matched against indexed dataset metadata using the BM25 retrieval model. The retrieved datasets are then evaluated using a fuzzy inference system that combines multiple metadata-driven relevance criteria.

The resulting relevance scores are used to generate the final ranking presented to the user. In addition, metadata-based explanations are generated to provide transparency regarding ranking decisions. This architecture supports both retrieval effectiveness and HCIR objectives.

4.4 Dataset Acquisition Layer

The first component of the framework is responsible for acquiring dataset metadata from the Swiss OGD portal, *opendata.swiss*. The portal provides access to datasets through a CKAN-based API, allowing metadata records to be retrieved programmatically.

Metadata fields collected from the portal include:

- Dataset title
- Dataset description
- Publisher information
- Tags and keywords
- Publication date
- Update date
- Dataset category
- Resource information

The acquired metadata forms the foundation of the retrieval process because relevance estimation is performed primarily on metadata rather than the full dataset contents.

4.5 Metadata Processing Layer

Dataset metadata obtained from the Swiss OGD portal undergoes preprocessing before indexing and retrieval. To emphasize the relative importance of metadata attributes, different weighting strategies are applied during document construction.

The title field receives the highest weight and is repeated three times within the document representation. Tags and groups are repeated twice to increase their influence during retrieval. Dataset descriptions are included once, while organizational information is only used as a fallback when structured metadata is unavailable.

This weighted representation enables important metadata attributes to contribute more strongly to retrieval performance while preserving the original metadata content.

The preprocessing pipeline includes:

1. Text normalization
2. Lowercase conversion
3. Removal of punctuation

4. Tokenization
5. Stop-word removal
6. Metadata validation

In addition to textual preprocessing, metadata completeness indicators are calculated. These indicators measure the availability of important metadata attributes and later contribute to fuzzy relevance estimation.

Metadata preprocessing improves retrieval effectiveness by reducing inconsistencies and ensuring that dataset descriptions can be compared more reliably during the retrieval process.

4.6 BM25 Retrieval Component

The first stage of relevance estimation is performed using the BM25 retrieval model. BM25 was selected because it provides robust relevance estimation, document length normalization, and strong retrieval performance across diverse IR tasks.

When a user submits a query, BM25 calculates similarity scores between the query and available dataset metadata. The resulting scores provide an initial ranking of candidate datasets.

The BM25 score is computed using:

$$BM25(D, Q) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, D)(k_1 + 1)}{f(q_i, D) + k_1 \left(1 - b + b \frac{|D|}{avgdl}\right)} \quad (4.1)$$

where the variables correspond to query term frequencies, document lengths, and inverse document frequency values.

The BM25 component serves as the retrieval foundation upon which fuzzy relevance estimation is subsequently applied.

4.7 Fuzzy Ranking Framework

While BM25 provides an effective estimate of textual similarity, relevance within OGD environments depends on additional factors. Consequently, a fuzzy inference system was developed to combine multiple metadata-driven relevance criteria into a unified relevance score.

The fuzzy ranking framework evaluates:

- BM25 similarity score
- Metadata completeness

- Dataset recency
- Resource availability

Each input variable is transformed into fuzzy linguistic values using membership functions. The fuzzy inference system then applies predefined rules to estimate overall relevance.

The output of the fuzzy system is a final relevance score that reflects both textual similarity and metadata quality characteristics. This approach enables more flexible and human-centered ranking behavior than traditional keyword-based retrieval alone.

4.8 Fuzzy Membership Functions

Membership functions define how numerical input variables are transformed into linguistic concepts used by the fuzzy inference engine. Depending on the characteristics of each variable, the implementation employs between three and five linguistic terms. For example, similarity is represented using five linguistic categories:

- Not Relevant,
- Somewhat Relevant
- Relevant
- Highly Relevant
- Exact Match

While dataset recency is represented using five temporal categories ranging from Very Recent to Very Old. This richer linguistic representation enables the fuzzy inference system to model gradual transitions in relevance more accurately than a simple three-level categorization.

4.9 Fuzzy Rule Base

The fuzzy rule base represents the core reasoning mechanism of the proposed retrieval framework. The rules combine multiple relevance criteria using IF-THEN statements that resemble human decision-making.

Examples of fuzzy rules include:

- IF BM25 Similarity is High AND Metadata Completeness is High AND Dataset Recency is High AND Resource Availability is High THEN Relevance is Very High.

- IF BM25 Similarity is Medium AND Metadata Completeness is High THEN Relevance is High.
- IF BM25 Similarity is Low THEN Relevance is Low.

The rule base enables the system to incorporate expert knowledge and domain-specific reasoning into the ranking process. Furthermore, the explicit structure of the rules supports transparency and explainability.

4.10 Explainability Component

Explainability is incorporated as a fundamental design requirement within the proposed framework. After relevance scores are generated, the system produces human-readable explanations that describe why datasets were ranked highly.

Explanations are derived from metadata attributes and fuzzy inference outcomes. For example, a highly ranked dataset may receive an explanation indicating that it achieved strong keyword similarity, contained complete metadata, and was recently updated.

The explainability component improves transparency and supports user trust by making ranking decisions understandable. This feature aligns closely with HCIR principles and distinguishes the proposed framework from traditional black-box retrieval systems.

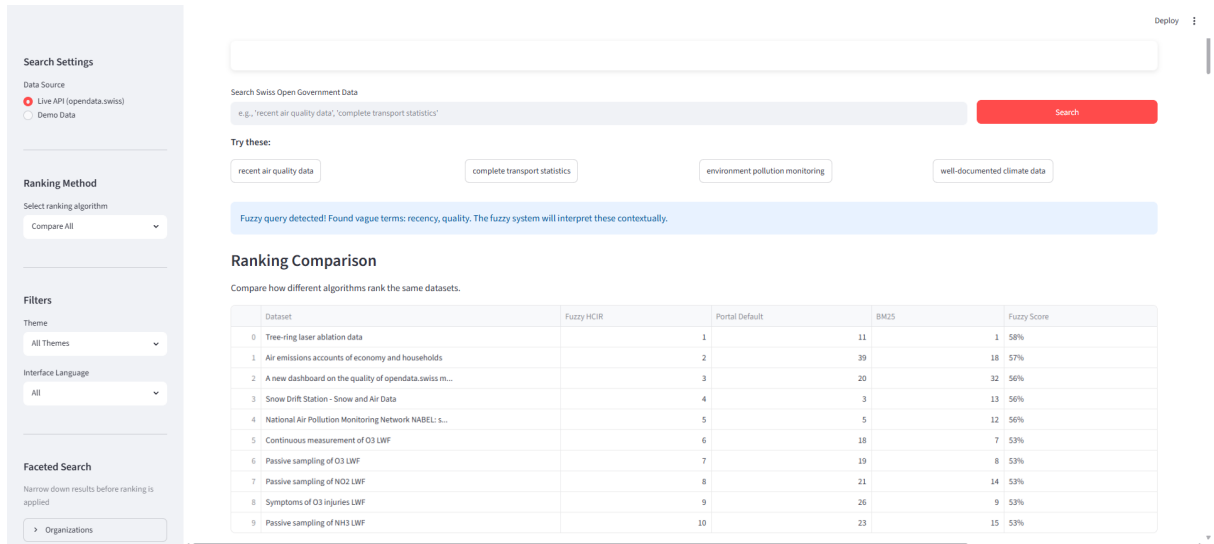


Figure 4.1: Comparison of dataset rankings generated by the proposed Fuzzy HCIR framework, BM25 retrieval, and the default portal ranking mechanism.

Figure 4.1 illustrates the ranking comparison interface developed within the proposed retrieval framework. The interface enables direct comparison between rankings produced by the Fuzzy HCIR approach, the BM25 retrieval model, and the default ranking mechanism provided by the opendata.swiss portal. By presenting ranking positions side by side,

users can observe how the incorporation of metadata completeness and dataset recency influences the final ordering of search results. This functionality improves transparency and allows users to better understand the impact of fuzzy relevance estimation compared with traditional keyword-based retrieval approaches. The comparison view therefore supports the HCIR objective of making ranking decisions more interpretable and understandable to users [3, 27].

4.11 User Interface Design

The UI was developed using Streamlit and serves as the primary interaction point between users and the retrieval system. The interface allows users to submit search queries, explore ranked datasets, and view generated explanations.

Key interface features include:

- Search query input
- Ranked dataset presentation
- Relevance score visualization
- Metadata summaries
- Explanation display

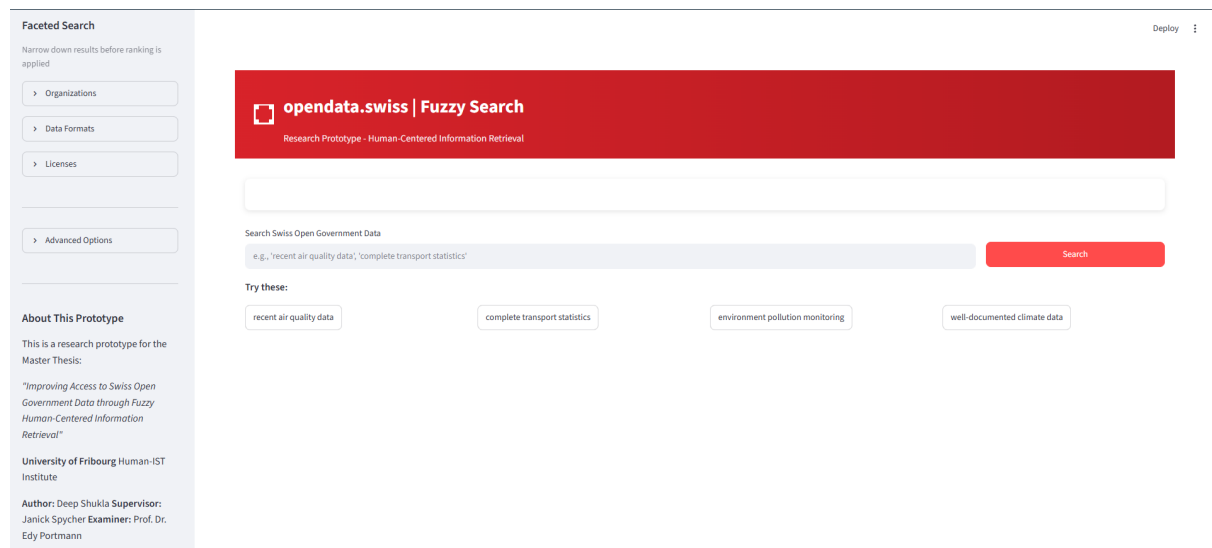


Figure 4.2: Main user interface of the proposed Fuzzy HCIR prototype for Swiss Open Government Data.

Figure 4.2 illustrates the main UI developed for the proposed retrieval framework. The design integrates query submission, faceted navigation, and metadata-based filtering

within a unified environment to support dataset discovery. The central search area enables users to express information needs using natural-language queries, while the faceted search panel provides mechanisms for narrowing result sets before ranking is applied. Additionally, example queries are provided to support exploratory search behaviour and reduce user effort during query formulation. The interface was designed according to HCIR principles, emphasizing usability, transparency, and user control throughout the search process [3, 26].

The interface was designed according to HCIR principles, emphasizing simplicity, transparency, and support for exploratory search activities.

4.12 System Workflow

The overall retrieval workflow can be summarized as follows:

1. User submits query.
2. Metadata is retrieved and preprocessed.
3. BM25 calculates initial similarity scores.
4. Fuzzy inference evaluates multiple relevance criteria.
5. Final relevance scores are generated.
6. Datasets are ranked.
7. Explanations are generated.
8. Results are displayed to the user.

This workflow integrates traditional retrieval techniques with fuzzy reasoning and explainability mechanisms to support HCIR objectives.

4.13 Chapter Summary

This chapter presented the architecture and design of the proposed HCIR framework. The system combines BM25 retrieval, fuzzy logic-based relevance estimation, and explainability mechanisms to support effective dataset discovery within OGD environments. The architecture was designed to balance retrieval effectiveness, transparency, interpretability, and user experience. The next chapter describes the implementation of the prototype and the technologies used to realize the proposed framework.

Chapter 5

Prototype Implementation

This chapter describes the implementation of the proposed HCIR framework. While the previous chapter presented the conceptual design of the system, this chapter focuses on the technologies, software components, and implementation procedures used to realize the proposed architecture. The implementation integrates OGD retrieval, BM25 ranking, fuzzy inference-based relevance estimation, and explainability mechanisms into a functional prototype designed to support dataset discovery.

5.1 Introduction

The implementation phase transforms the conceptual design presented in the previous chapter into a working software artifact. The primary objective of the implementation is to develop a prototype capable of retrieving datasets from the Swiss OGD portal, ranking them using fuzzy logic-based relevance estimation, and providing explainable retrieval results.

The implementation follows the DSR process and emphasizes practicality, transparency, and reproducibility. The resulting prototype serves as the evaluation artifact through which the research questions are investigated and validated.

5.2 Development Environment and Technology Stack

The prototype was implemented using Python and supporting open-source libraries selected according to the functional requirements of the system. Table 5.1 summarizes the primary technologies used during development.

Table 5.1: Technology Stack

Component	Technology
Programming Language	Python
User Interface	Streamlit
Data Processing	Pandas, NumPy
Scientific Computing	SciPy
Visualization	Matplotlib, Seaborn
API Communication	Requests
PDF Processing	PyPDF2
Testing	Pytest, Pytest-Cov
Code Quality	Black, Flake8, MyPy

5.2.1 Custom Implementation Strategy

A deliberate design decision was made to implement the BM25 retrieval model and fuzzy inference engine from scratch rather than relying on external libraries. Although several mature implementations are available, custom development provided greater control over the retrieval pipeline and allowed complete visibility into each stage of relevance estimation.

The BM25 ranking component was implemented directly using the probabilistic ranking formulation proposed by Robertson and Walker [34]. The implementation includes inverse document frequency calculation, document length normalization, and configurable ranking parameters. This approach ensured that all retrieval computations remained transparent and could be adapted to the specific characteristics of OGD metadata.

Similarly, the fuzzy inference engine was implemented as a custom Mamdani-style system supporting multiple membership function types, configurable fuzzy operators, rule evaluation mechanisms, aggregation procedures, and centroid-based defuzzification. Implementing these components directly enabled detailed experimentation with different fuzzy configurations and facilitated the generation of explainable ranking decisions.

The custom implementation strategy contributes to the transparency and interpretability objectives of the research by providing complete control over the ranking process and eliminating dependencies on external black-box libraries.

5.3 Dataset Acquisition Implementation

The dataset acquisition component was responsible for collecting metadata from the Swiss OGD portal. The implementation utilized the CKAN API provided by the *opendata.swiss* platform to access publicly available dataset metadata [11, 10].

The acquisition process retrieved essential metadata attributes including dataset titles, descriptions, tags, groups, organizational information, publication dates, and update timestamps. These metadata elements formed the foundation of the retrieval and ranking framework developed in this research. The selection of metadata attributes was guided by previous research highlighting the importance of metadata quality and completeness for dataset discoverability and retrieval effectiveness [21, 6].

To improve robustness, the acquisition module incorporated validation procedures to handle incomplete or inconsistent metadata records. Retrieved datasets were subsequently transformed into a structured format suitable for metadata processing, indexing, and retrieval operations. Such preprocessing is particularly important in OGD environments where metadata quality may vary across publishers [28, 1].

The acquisition layer was designed to support periodic metadata updates, ensuring that newly published datasets and metadata modifications could be incorporated into the retrieval framework without requiring major system modifications. This capability helps maintain dataset freshness and improves the long-term usability of the retrieval system.

Evaluation Corpus. Although the *opendata.swiss* portal contains substantially more datasets, the experimental evaluation presented in this thesis was conducted using a curated corpus of 500 datasetsg retrieved through the CKAN API. The corpus consisted primarily of datasets published by the Swiss Federal Statistical Office (FSO), the Canton of Thurgau, and the Swiss National Library. The subset was selected to provide a practical dataset for prototype development and evaluation while maintaining realistic metadata characteristics. Therefore, the reported findings should be interpreted within the scope of this evaluation corpus and should not be generalized to the entire *opendata.swiss* repository without further large-scale validation.

5.4 Metadata Weighting Strategy

The retrieval framework employs a metadata weighting strategy to emphasize the relative importance of different metadata attributes during document construction. Rather than treating all metadata fields equally, the implementation assigns greater importance to fields that are expected to contribute more significantly to dataset discoverability and relevance estimation. Previous IR research has demonstrated that the effectiveness of retrieval models can be improved by appropriately weighting informative document attributes [35, 34].

Dataset titles are considered the most informative metadata element and therefore receive the highest weighting. Tags and groups are also assigned increased importance because they provide structured thematic information that can improve retrieval effectiveness. Descriptions are included to provide contextual information regarding dataset

content, while organizational information is used only as a fallback when structured metadata attributes are unavailable.

The implemented weighting strategy is summarized as follows:

- Title: weight 3
- Tags: weight 2
- Groups: weight 2
- Description: weight 1
- Organization: fallback attribute

During document construction, title information is repeated three times, tags and groups are repeated twice, and descriptions are included once. This weighted representation increases the influence of highly informative metadata attributes while preserving all available metadata content.

The weighting strategy was specifically designed to improve thematic matching between user queries and dataset metadata. By emphasizing metadata fields that are most likely to reflect dataset topics and user intent, the framework supports more effective dataset discovery within the Swiss OGD environment. This design is consistent with established IR principles that prioritize informative content when estimating document relevance [34, 35].

5.5 Custom BM25 Implementation

The retrieval component was implemented using a custom BM25 algorithm specifically developed for the proposed HCIR framework. Although existing BM25 libraries are available, a custom implementation was selected to provide complete control over metadata processing, term weighting, similarity estimation, and integration with the fuzzy ranking framework. This approach also ensured transparency and allowed the retrieval process to be adapted to the specific characteristics of OGD metadata [34].

The implementation follows the probabilistic relevance framework proposed by Robertson and Zaragoza [34]. BM25 was selected because it provides effective handling of term frequency, document length normalization, and relevance estimation while maintaining interpretability and computational efficiency. Compared with traditional term-weighting approaches, BM25 offers a more robust mechanism for ranking documents and has consistently demonstrated strong retrieval performance across IR applications [34, 35].

The inverse document frequency (IDF) used within the implementation is calculated as:

$$IDF = \log \left(\frac{N - df + 0.5}{df + 0.5} + 1 \right) \quad (5.1)$$

where (N) represents the total number of indexed datasets and (df) denotes the document frequency of a query term. This formulation reduces the influence of highly frequent terms while assigning greater importance to terms that are more discriminative within the dataset collection [34].

The implementation employs the standard BM25 parameters $k_1 = 1.5$ and $b = 0.75$, where k_1 represents the term frequency saturation parameter and b represents the document length normalization parameter. The parameter k_1 controls the extent to which repeated occurrences of a query term contribute to the relevance score, while b regulates the influence of document length normalization. These parameter values are widely adopted within IR research and provide a balanced trade-off between retrieval effectiveness and computational efficiency [34].

Unlike traditional document retrieval systems that primarily operate on unstructured text, the proposed implementation applies BM25 to weighted metadata representations constructed from dataset titles, tags, groups, descriptions, and organizational information. This adaptation enables BM25 to operate effectively within the OGD environment while preserving compatibility with the subsequent fuzzy ranking process.

The custom implementation further facilitates integration with metadata completeness assessment, dataset recency evaluation, explainability generation, and fuzzy inference mechanisms. Consequently, BM25 serves as the foundation for thematic similarity estimation within the overall retrieval framework and provides the primary input to the fuzzy ranking component developed in this research.

5.6 Similarity Normalization

The raw BM25 scores generated by the retrieval component are not directly supplied to the fuzzy inference engine. Instead, a normalization step is performed to transform the scores into a bounded similarity measure suitable for fuzzy processing.

The normalization process combines the BM25 score with query coverage information. Query coverage represents the proportion of query terms successfully matched within the metadata representation of a dataset. By incorporating coverage information, the framework rewards datasets that match a larger portion of the user’s query while maintaining the relevance estimation provided by BM25.

The normalized similarity value is calculated as:

$$Similarity = \frac{Score}{Score + 1.0} \times Coverage \quad (5.2)$$

where (Score) represents the raw BM25 retrieval score and (Coverage) represents the proportion of matched query terms. This transformation constrains similarity values to the interval $[0,1]$, making them suitable inputs for the fuzzy inference system.

The normalization process also reduces the influence of extremely large BM25 values that may occur for highly relevant datasets. Consequently, the fuzzy ranking component receives consistent and interpretable similarity values that can be combined with metadata completeness and dataset recency indicators.

The resulting normalized similarity score serves as the thematic similarity input of the fuzzy ranking framework. This design enables the integration of traditional IR techniques with fuzzy logic-based relevance estimation while maintaining transparency and interpretability throughout the ranking process [34, 42, 4].

5.7 Custom Mamdani Fuzzy Inference Engine

The fuzzy ranking component was implemented using a custom Mamdani fuzzy inference engine specifically developed for the requirements of the proposed HCIR framework. Although several fuzzy logic libraries are available, a custom implementation was selected to provide full control over membership functions, rule evaluation, inference procedures, and explainability generation.

The implementation follows the Mamdani fuzzy inference approach, one of the most widely adopted fuzzy reasoning models due to its interpretability and ability to represent human decision-making processes through linguistic rules [24, 42]. This characteristic aligns closely with the objectives of HCIR, where transparency and user understanding are essential.

The fuzzy inference engine consists of four primary stages:

1. Fuzzification
2. Rule Evaluation
3. Aggregation
4. Defuzzification

During fuzzification, numerical inputs are transformed into linguistic variables using predefined membership functions. Rule evaluation applies the fuzzy rule base to determine the degree of activation for each rule. Aggregation combines the outputs of all activated rules, while defuzzification converts the aggregated fuzzy output into a single numerical relevance score.

The custom implementation supports multiple membership function types, including:

- Triangular
- Trapezoidal
- Gaussian
- Sigmoid
- Generalized Bell

Supporting multiple membership functions provides flexibility during system design and experimentation while allowing the fuzzy ranking framework to adapt to different relevance estimation requirements.

The final output generated by the Mamdani inference engine represents the overall relevance score of a dataset. The inference process integrates four complementary relevance indicators: thematic similarity, metadata completeness, dataset recency, and resource availability. Combining these indicators through fuzzy reasoning enables the framework to model gradual relevance relationships while simultaneously considering both topical relevance and important metadata characteristics. The resulting relevance score forms the basis for the final dataset ranking and supports transparent, human-interpretable decision-making, which is consistent with the principles of HCIR and explainable decision support systems [4, 42, 3, 27].

The use of a custom Mamdani inference engine also improves transparency because every ranking decision can be traced back to specific membership functions and fuzzy rules. This capability directly supports the explainability objectives of the proposed HCIR framework and contributes to greater user trust in ranking outcomes [3, 27].

Justification for Selecting the Mamdani Fuzzy Inference System

The selection of the Mamdani fuzzy inference system was guided primarily by the objective of supporting HCIR through transparent and interpretable ranking decisions. Unlike Takagi–Sugeno (TSK) fuzzy systems, which generate mathematical output functions, Mamdani inference represents both inputs and outputs using linguistic variables such as *Low*, *Medium*, and *High*. This linguistic representation enables ranking decisions to be communicated in a manner that is understandable to end users.

Since one of the primary objectives of this research is to improve transparency and explainability, interpretability was considered more important than computational efficiency. The linguistic outputs generated by the Mamdani approach integrate naturally with the explanation component of the proposed framework, allowing users to understand how metadata completeness, keyword relevance, publication recency, and organizational quality jointly influence dataset ranking.

Although Takagi–Sugeno fuzzy systems are widely used in control and optimization applications because of their computational efficiency, the Mamdani approach is generally considered more suitable for decision-support systems in which human interpretation of inference results is required. Consequently, the Mamdani fuzzy inference system was selected as the most appropriate reasoning mechanism for the proposed HCIR framework [42, 38, 13].

Why Learning-to-Rank Was Not Adopted

Learning-to-Rank (LTR) approaches represent an important class of IR algorithms that learn ranking functions directly from labelled training data. Although these techniques have demonstrated strong retrieval performance in many large-scale retrieval tasks, they were not adopted in this research for several reasons.

First, the benchmark developed for this study contains a relatively small number of representative queries and manually reviewed relevance judgments, which is insufficient for training and validating a supervised Learning-to-Rank model. Second, many Learning-to-Rank algorithms operate as complex machine learning models whose ranking decisions are difficult to interpret. Since the objective of this research is to support HCIR through transparent and explainable ranking, interpretability was considered a primary design requirement.

Consequently, a rule-based Mamdani fuzzy inference system was selected because it enables explicit modelling of multiple metadata-based relevance criteria while preserving transparency and allowing users to understand how ranking decisions are generated [42, 9].

Design rationale. The proposed fuzzy inference system combines data-calibrated membership functions with an expert-defined rule base because these two components represent different types of knowledge. Membership functions describe observable characteristics of the metadata, such as similarity scores, metadata completeness, recency, and resource availability. Since these variables are directly measurable from the dataset collection, their breakpoints were calibrated using the statistical distribution of the corpus to ensure that the linguistic categories accurately reflected the characteristics of the available metadata.

In contrast, the fuzzy rules represent human judgement regarding how different relevance factors should be combined during ranking. Such preferences cannot be derived reliably from the relatively small benchmark query set used in this study and therefore require expert knowledge. Defining the rules manually also preserves transparency because every ranking decision can be traced back to explicit IF–THEN rules that are understandable to both developers and users. This combination of data-driven calibration and expert knowledge provides a balance between empirical adaptation and interpretability, consistent with the objectives of HCIR.

5.7.1 Fuzzy Ranking Inputs

The proposed fuzzy ranking framework utilizes four complementary relevance indicators to evaluate dataset quality and relevance. These indicators were selected to address limitations commonly observed in traditional retrieval systems that rely exclusively on textual similarity measures. By incorporating multiple relevance factors, the framework provides a more comprehensive and human-centered assessment of dataset usefulness [3, 4].

The first input is **Thematic Similarity**, which represents the degree of correspondence between the user’s query and the metadata content of a dataset. Thematic similarity is generated by the BM25 retrieval component and subsequently normalized before being supplied to the fuzzy inference engine. This indicator captures the topical relevance of datasets with respect to user information needs [34].

The second input is **Metadata Completeness**. This indicator evaluates the extent to which important metadata attributes are available for a dataset. Datasets containing complete and well-structured metadata generally provide greater contextual information and are easier for users to interpret and assess. Previous studies have identified metadata quality as a critical factor influencing dataset discoverability and reuse within OGD environments [21, 6, 28].

The third input is **Dataset Recency**. Recency measures the freshness of a dataset based on available publication and update information. In many OGD scenarios, users prefer datasets that have been recently updated because they are more likely to reflect current conditions and provide reliable information. Incorporating recency therefore enables the ranking framework to account for temporal relevance in addition to thematic relevance [16, 1].

The fourth input is **Resource Availability**. This indicator evaluates the practical accessibility and usability of a dataset by considering the availability of associated resources, such as downloadable files or supporting data services. Even when a dataset is topically relevant, limited resource availability may reduce its usefulness for users wishing to inspect or reuse the data. Incorporating this criterion therefore extends the ranking process beyond topical relevance by accounting for the practical utility of search results, an important consideration in HCIR and OGD environments [1, 3].

Before fuzzy inference is performed, each input variable is transformed into fuzzy linguistic variables through the fuzzification process. This transformation allows quantitative measurements to be represented using human-interpretable concepts such as *Low*, *Medium*, and *High*. The resulting fuzzy representations provide the foundation for rule-based reasoning and relevance estimation within the fuzzy ranking framework [42, 24].

By combining thematic similarity, metadata completeness, dataset recency, and resource availability, the proposed framework evaluates dataset relevance from multiple

complementary perspectives rather than relying solely on textual matching. This multi-criteria approach better reflects how users assess dataset usefulness during exploratory search and supports the objectives of HCIR by integrating relevance, metadata quality, temporal validity, and practical accessibility into a single interpretable ranking process [3, 26, 4].

5.7.2 Explanation Generation

A key objective of the proposed HCIR framework is to improve transparency and user understanding during the dataset discovery process. Traditional retrieval systems typically present ranked results without explaining the factors that influenced ranking decisions. As a result, users may find it difficult to understand why particular datasets appear at the top of search results. Explainability addresses this limitation by providing meaningful descriptions of the reasoning process underlying relevance estimation [27, 3].

The explanation generation component was designed to transform fuzzy reasoning outcomes into human-readable descriptions. Rather than presenting only numerical relevance scores, the system identifies the primary factors contributing to dataset rankings and communicates these factors through concise textual explanations.

Explanations are generated directly from the outputs of the ranking process rather than from predefined textual templates. For each ranked dataset, the system identifies the relevance indicators that contributed most strongly to the final score, including thematic similarity, metadata completeness, dataset recency, and resource availability. These values are combined with information derived from the fuzzy inference process to produce faithful explanations that accurately reflect the factors influencing the ranking decision. Consequently, the explanation component communicates the actual reasoning underlying the computed relevance score rather than presenting a generic post-hoc justification, thereby improving transparency and user trust [27, 9, 3].

The explanation generation process follows three primary steps:

1. Identification of dominant ranking factors.
2. Analysis of fuzzy inference outcomes.
3. Generation of human-readable explanations.

This approach enables users to understand how different relevance indicators contributed to ranking decisions. By exposing the reasoning process, the system promotes transparency and supports informed decision-making during exploratory search activities.

The explanation module is tightly integrated with the ranking pipeline rather than operating as an independent post-processing component. After the fuzzy inference engine

computes the final relevance score, the system records the contribution of each ranking factor, including thematic similarity, metadata completeness, dataset recency, and resource availability. These recorded values are subsequently transformed into human-readable explanations presented alongside the ranked results. Because the explanations are derived directly from the same metadata features and fuzzy reasoning process used for ranking, they faithfully represent the underlying decision-making process rather than providing approximate or post-hoc justifications.

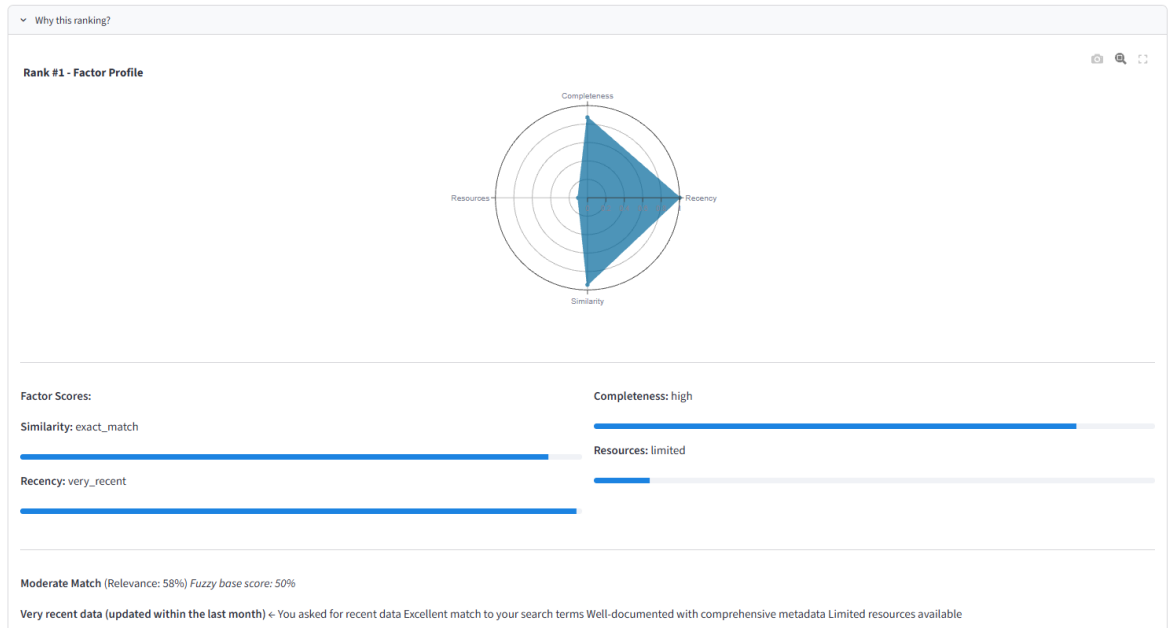


Figure 5.1: Explanation interface illustrating the factors contributing to dataset relevance, including thematic similarity, metadata completeness, recency, and resource availability.

Figure 5.1 illustrates the explanation interface developed to improve transparency within the proposed HCIR framework. For each ranked dataset, the system presents a detailed breakdown of the factors contributing to the final relevance score, including thematic similarity, metadata completeness, dataset recency, and resource availability. A visual radar chart summarizes the relative contribution of each factor, while textual explanations provide additional context regarding the ranking decision. By exposing the reasoning process behind relevance estimation, the explanation component enables users to better understand why particular datasets appear at specific ranking positions. This functionality supports transparency, interpretability, and user trust, which are central objectives of HCIR and explainable information systems [3, 27, 42].

The use of fuzzy logic further enhances explainability because the underlying reasoning process is based on linguistic variables and interpretable rule structures. Unlike many complex machine learning models that operate as black boxes, fuzzy inference systems provide clear relationships between inputs, rules, and outputs [42, 4]. Consequently,

users can better understand how relevance scores are produced and how different dataset characteristics influence ranking outcomes.

From a HCIR perspective, explainability contributes to user trust, system transparency, and improved interaction quality. Providing explanations alongside search results enables users to evaluate dataset relevance more effectively and supports the broader objective of making OGD more accessible and understandable [3, 27].

5.8 User Interface Implementation

The UI was implemented using Streamlit to provide an interactive and user-friendly environment for dataset discovery and exploration. Streamlit was selected because it supports rapid development of web-based applications while enabling seamless integration with Python-based retrieval and ranking components.

The interface was designed according to the principles of HCIR, emphasizing usability, transparency, and user understanding throughout the search process [3, 26]. Users can submit natural-language queries, inspect ranked datasets, and access explanations describing the factors that influenced ranking decisions.

The search interface consists of several interconnected components. The query input component allows users to enter information needs using natural language expressions. Submitted queries are processed by the retrieval framework, which generates thematic similarity estimates and subsequently applies fuzzy relevance estimation.

Search results are presented in ranked order according to the relevance scores generated by the fuzzy inference engine. For each dataset, the interface displays key metadata attributes including titles, descriptions, tags, groups, and organizational information. Presenting metadata directly within the interface enables users to quickly evaluate dataset relevance without requiring additional navigation.

In addition to ranked results, the interface provides explanation panels that communicate the primary factors contributing to relevance estimation. These explanations are generated from the fuzzy reasoning process and help users understand how thematic similarity, metadata completeness, and dataset recency influenced ranking outcomes. Providing such explanations improves transparency and supports informed decision-making during exploratory search activities [27, 3].

The interface also supports exploratory dataset discovery by enabling users to compare datasets, examine metadata characteristics, and refine search strategies. This functionality aligns with HCIR principles that view information seeking as an interactive and evolving process rather than a simple query-response interaction [26].

Overall, the UI serves as the primary interaction point between users and the retrieval framework, integrating retrieval, fuzzy ranking, and explainability components into a unified environment for OGD discovery.

5.9 Integration of Components

The proposed HCIR framework consists of multiple interconnected components that operate together to support dataset discovery within the Swiss OGD environment. The integration process combines metadata acquisition, BM25 retrieval, similarity normalization, fuzzy relevance estimation, explanation generation, and user interaction within a unified architecture.

The retrieval process begins when a user submits a query through the interface. The query is forwarded to the retrieval component, where BM25 is applied to estimate thematic similarity between the query and available dataset metadata. The resulting similarity scores provide an initial assessment of dataset relevance based on textual matching principles [34].

Following retrieval, the similarity normalization component transforms raw BM25 scores into bounded similarity values suitable for fuzzy processing. The normalized thematic similarity score is subsequently combined with metadata completeness and dataset recency indicators to create the input variables required by the fuzzy inference engine.

The fuzzy ranking component evaluates these inputs through fuzzification, rule evaluation, aggregation, and defuzzification. This process produces a final relevance score that reflects multiple dimensions of dataset quality and relevance rather than relying solely on textual similarity [42, 24, 4].

The explanation generation component operates in parallel with the ranking process. Information obtained from the fuzzy inference engine is analyzed to identify the dominant factors contributing to ranking decisions. These factors are subsequently transformed into human-readable explanations that accompany search results and improve transparency [27].

Finally, the UI presents ranked datasets together with relevant metadata and generated explanations. Users can inspect search results, compare datasets, and better understand the reasoning process underlying relevance estimation. This integrated workflow supports the objectives of HCIR by combining retrieval effectiveness, explainability, and user-centered interaction within a single framework [3, 26].

5.10 Testing and Validation

Testing and validation activities were conducted throughout the development process to ensure the reliability, correctness, and usability of the proposed HCIR framework. The objective of testing was to verify that individual components functioned correctly and that the integrated system produced consistent and meaningful retrieval results.

Several levels of testing were performed, including functional testing, integration testing, retrieval validation, fuzzy inference validation, and UI testing. This multi-level ap-

proach ensured that both individual components and the complete framework operated as intended.

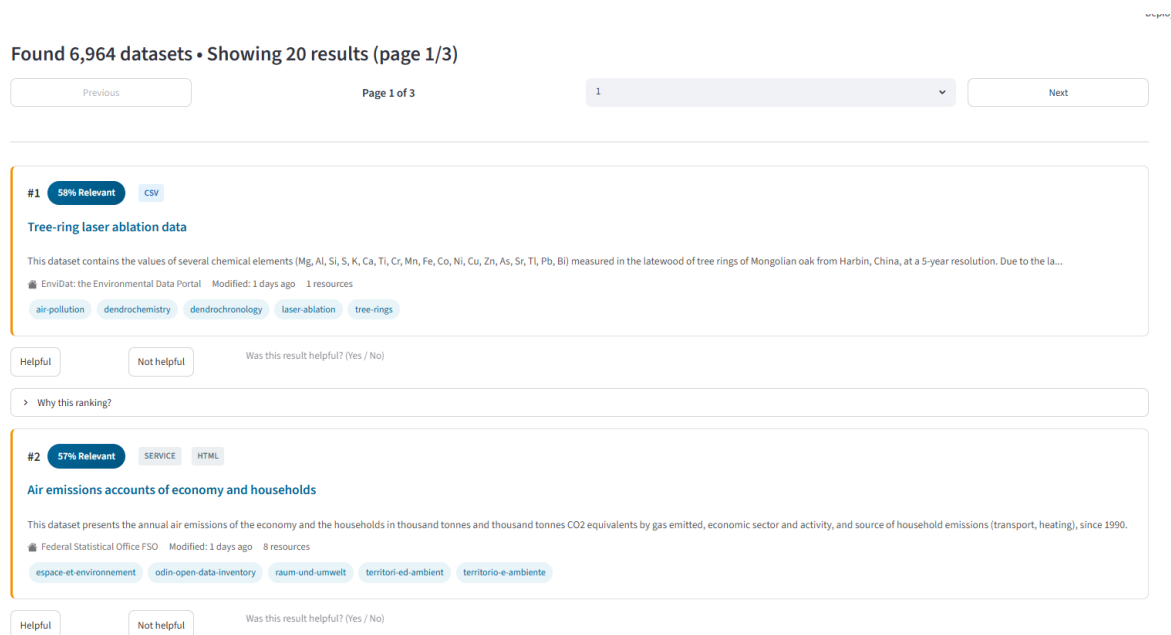


Figure 5.2: Search results interface displaying ranked datasets, relevance scores, metadata attributes, and access to explanation functionality.

Figure 5.2 presents the search results interface of the proposed Fuzzy HCIR system. Retrieved datasets are displayed in ranked order according to the relevance scores generated by the fuzzy inference engine. Each result includes key metadata attributes such as dataset title, description, publisher information, update date, resource count, and thematic tags. The interface also provides relevance indicators and direct access to explanation functionality, enabling users to understand why a dataset was assigned a particular ranking position. By combining ranked retrieval with transparent presentation of metadata and relevance information, the interface supports effective dataset exploration and aligns with HCIR principles that emphasize usability, transparency, and informed decision-making [3, 26, 27].

5.10.1 Functional Testing

Functional testing was conducted to verify the behavior of individual system components. The dataset acquisition module, metadata processing component, BM25 retrieval engine, fuzzy inference engine, and explanation generation module were tested independently to confirm correct operation under various input conditions.

Particular attention was given to validating metadata processing procedures, similarity calculations, fuzzy rule evaluation, and explanation generation mechanisms. Test cases were designed to verify that each component produced expected outputs and handled incomplete or inconsistent metadata appropriately.

5.10.2 Integration Testing

Integration testing was performed to verify communication between system components. The complete workflow, beginning with user query submission and ending with ranked dataset presentation, was evaluated to ensure seamless interaction among retrieval, ranking, explanation, and UI modules.

Integration testing confirmed that similarity values generated by the BM25 retrieval component were correctly transferred to the fuzzy inference engine and that final relevance scores were successfully communicated to the explanation and visualization components.

5.10.3 Retrieval Validation

Retrieval validation focused on assessing whether the BM25 component produced meaningful thematic similarity estimates. A variety of test queries were submitted to the system, and the resulting rankings were manually inspected to determine whether highly relevant datasets appeared near the top of the ranking list.

The validation process demonstrated that the retrieval component effectively identified datasets whose metadata closely matched user information needs. These results provided confidence that thematic similarity estimates could serve as reliable inputs to the fuzzy ranking framework [34].

5.10.4 Fuzzy Inference Validation

Validation of the fuzzy inference engine focused on verifying the correctness of fuzzification, rule evaluation, aggregation, and defuzzification procedures. Test scenarios representing different combinations of thematic similarity, metadata completeness, and dataset recency were evaluated to ensure that the resulting relevance scores were consistent with the intended rule base.

The validation process confirmed that the fuzzy ranking framework produced interpretable relevance estimates and appropriately reflected the influence of multiple ranking factors. These findings are consistent with established principles of fuzzy reasoning and fuzzy IR [42, 24, 4].

5.10.5 User Interface Testing

UI testing was conducted to verify usability, navigation, and interaction functionality. The search interface, ranking display, metadata presentation, and explanation components were evaluated to ensure that users could successfully perform search and exploration tasks.

Testing confirmed that the interface supported efficient query submission, result inspection, and explanation interpretation. The interface therefore provided an appropriate

environment for subsequent user evaluation activities conducted during the assessment phase of the research.

Overall, the testing and validation activities demonstrated that the proposed framework operated reliably and that all major components functioned correctly both individually and as part of the integrated system.

5.11 Implementation Challenges

The development of the proposed HCIR framework involved several technical and methodological challenges. These challenges emerged during metadata acquisition, retrieval model implementation, fuzzy system design, and explainability generation. Addressing these challenges was necessary to ensure the reliability, effectiveness, and usability of the final prototype.

5.11.1 Metadata Quality and Consistency

One of the primary challenges involved the quality and consistency of metadata available through the Swiss OGD portal. Dataset publishers often provide metadata with varying levels of completeness, detail, and standardization. Some datasets contained comprehensive metadata descriptions, while others lacked important attributes such as tags, groups, or detailed descriptions.

This inconsistency complicated both retrieval and ranking processes because metadata quality directly influences dataset discoverability. To mitigate this issue, preprocessing and validation procedures were incorporated into the acquisition pipeline to improve metadata handling and ensure that incomplete records could still be processed effectively [1, 21, 28].

5.11.2 Balancing Multiple Relevance Factors

Traditional retrieval systems typically rely on textual similarity as the primary ranking criterion. In contrast, the proposed framework integrates thematic similarity, metadata completeness, and dataset recency and resource availability. Determining how these factors should interact during relevance estimation represented a significant design challenge.

The fuzzy inference approach was selected to address this issue because it provides a flexible mechanism for combining multiple relevance indicators while preserving interpretability. Designing an appropriate rule base required iterative refinement to ensure that ranking outcomes reflected realistic user expectations and relevance judgments [42, 24, 4].

5.11.3 Design of Membership Functions

Another challenge involved the definition of membership functions used during fuzzification. Membership functions determine how numerical inputs are transformed into linguistic concepts such as *Low*, *Medium*, and *High*. Poorly defined membership functions can lead to unrealistic relevance estimations and unstable ranking behavior.

Several membership function configurations were evaluated during development to identify representations that produced meaningful and interpretable ranking outcomes. The final configuration was selected based on consistency, interpretability, and compatibility with the retrieval objectives of the framework [43, 42].

5.11.4 Explainability and User Understanding

Providing meaningful explanations for ranking decisions presented an additional challenge. While numerical relevance scores can indicate dataset importance, they do not necessarily help users understand why particular datasets were ranked highly.

To address this limitation, the explanation generation component was designed to identify dominant ranking factors and translate them into human-readable descriptions. This approach improves transparency and aligns with HCIR principles that emphasize user understanding and trust [27, 3].

5.11.5 System Integration

Integrating multiple components into a unified retrieval framework also required careful coordination. Metadata acquisition, BM25 retrieval, similarity normalization, fuzzy inference, explanation generation, and UI components needed to communicate efficiently while maintaining consistent data representations.

Ensuring compatibility among these components required iterative testing and refinement throughout the development process. Successful integration was essential for achieving the overall objective of providing an effective and explainable dataset retrieval environment.

Despite these challenges, the final implementation successfully integrated all major components into a functional prototype capable of supporting HCIR for Swiss OGD.

5.12 Chapter Summary

This chapter presented the implementation of the proposed HCIR framework for Swiss OGD. The implementation process included dataset acquisition, metadata processing, metadata weighting, custom BM25 retrieval, similarity normalization, fuzzy relevance estimation, explanation generation, and UI development.

A custom BM25 retrieval component was implemented to estimate thematic similarity between user queries and dataset metadata. The resulting similarity scores were normalized and combined with metadata completeness, dataset recency, and resource availability indicators through a custom Mamdani fuzzy inference engine. This approach enabled the framework to evaluate dataset relevance using multiple criteria while maintaining transparency and interpretability [34, 42, 24, 4].

The chapter further described the implementation of the explanation generation component and the Streamlit-based UI, both of which support the HCIR objectives of transparency, explainability, and user understanding [3, 27].

Finally, testing and validation activities confirmed the correct operation of individual components and the integrated system. The implementation challenges encountered during development were also discussed, together with the strategies adopted to address them.

Having presented the design and implementation of the proposed framework, the next chapter evaluates the system through experimental assessment and user-based evaluation to determine its effectiveness in supporting dataset discovery and retrieval within the Swiss OGD environment.

Chapter 6

Evaluation Methodology

This chapter presents the methodology used to evaluate the proposed HCIR framework. The objective of the evaluation is to determine whether the system effectively improves dataset discoverability while supporting transparency, explainability, and user-centered interaction. The chapter describes the evaluation objectives, participant selection process, data collection methods, evaluation metrics, and analysis procedures used to assess the performance and usability of the proposed retrieval framework.

This chapter presents the methodology used to evaluate the proposed HCIR framework. The objective of the evaluation is to determine whether the system effectively improves dataset discoverability while supporting transparency, explainability, and user-centered interaction. The chapter describes the evaluation objectives, participant selection process, data collection methods, evaluation metrics, and analysis procedures used to assess the performance and usability of the proposed retrieval framework.

6.1 Evaluation Objectives

The evaluation was designed to assess the extent to which the proposed retrieval framework achieves the objectives established in Chapter 1. Specifically, the evaluation sought to determine whether the integration of BM25 retrieval, fuzzy logic-based ranking, and explainability mechanisms contributes to improved dataset discovery and user experience.

The evaluation objectives are summarized as follows:

1. Assess the effectiveness of the retrieval framework in identifying relevant datasets.
2. Evaluate user perceptions of system usability.
3. Measure the perceived transparency of ranking decisions.
4. Assess the usefulness of generated explanations.
5. Investigate the contribution of fuzzy logic to HCIR.

6. Identify strengths and limitations of the proposed framework.

6.2 Evaluation Strategy

The proposed HCIR framework was evaluated using a mixed-method evaluation strategy consisting of two complementary components: (1) a quantitative IR evaluation and (2) a qualitative user-centered evaluation. This mixed-method approach is consistent with the principles of DSR, which recommend evaluating both the technical performance of an artifact and its usefulness for end users [14, 32].

The quantitative evaluation was performed using a frozen metadata snapshot of the Swiss OGD portal to ensure reproducibility. All retrieval systems were evaluated on the same corpus and benchmark query set under identical experimental conditions. To minimise assessment bias, a pooled evaluation methodology was adopted in which the top-ranked results returned by each retrieval system were combined and deduplicated before relevance assessment. Two independent judges assigned graded relevance labels (0 = not relevant, 1 = partially relevant, 2 = highly relevant), and all disagreements were resolved through adjudication to produce the final ground truth. Inter-rater agreement was measured using quadratic-weighted Cohen’s kappa to assess annotation consistency [17, 25].

Retrieval effectiveness was evaluated using Mean Average Precision (MAP), Precision at five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). These complementary metrics evaluate different aspects of retrieval quality, including overall ranking effectiveness, early precision, graded relevance, and the position of the first relevant result [17, 25].

To determine whether observed differences between retrieval systems were statistically meaningful rather than attributable to random variation, pairwise comparisons were performed using the Wilcoxon signed-rank test. Since multiple pairwise comparisons were conducted, Holm–Bonferroni correction was applied to control the family-wise error rate while maintaining greater statistical power than the standard Bonferroni correction [15]. In addition, non-parametric bootstrap confidence intervals were computed for each evaluation metric to quantify the uncertainty associated with the reported mean performance values. Finally, a sensitivity analysis was conducted to evaluate the robustness of the fuzzy inference system by perturbing the rule configuration and observing the resulting changes in retrieval effectiveness. Together, these analyses provide complementary evidence regarding the reliability and stability of the proposed HCIR framework beyond point estimates alone [8].

The robustness of the proposed fuzzy ranking framework was further investigated through a sensitivity analysis. Membership-function breakpoints were systematically perturbed and leave-one-rule-out experiments were conducted to evaluate the stability

of the resulting rankings and to assess the influence of individual fuzzy rules on retrieval performance.

The qualitative evaluation complemented the quantitative analysis by examining user perceptions of usability, transparency, explainability, and trust through controlled user sessions involving Master’s students performing representative OGD retrieval tasks. Together, the quantitative and qualitative evaluations provide a comprehensive assessment of both retrieval performance and the extent to which the proposed framework satisfies the objectives of HCIR [3, 26, 40].

6.3 Participants

The evaluation involved participants with varying levels of familiarity with IR systems, OGD, and data analysis. Including participants from diverse backgrounds was important because OGD portals are intended for a broad range of users, including researchers, students, policymakers, journalists, and citizens.

The participant group was intentionally selected because Master’s students and researchers regularly perform exploratory IR tasks and possess sufficient technical literacy to critically evaluate retrieval quality, explanation mechanisms, and system transparency. As the objective of this study was to conduct a formative evaluation of a research prototype rather than a large-scale usability validation, this participant group provided appropriate feedback on the proposed HCIR framework. Nevertheless, the findings should not be interpreted as representative of all OGD users, and future studies should include a broader range of stakeholders, including public administrators, journalists, policymakers, and citizens.

The participant group was selected using purposive sampling because the study aimed to obtain informed feedback from individuals capable of critically evaluating IR systems and OGD search tasks. Master’s students in Computer Science were considered appropriate participants because they possess sufficient technical knowledge to understand retrieval behaviour while still representing prospective end users of data discovery platforms.

The purpose of the user study was formative rather than summative. Consequently, the results are intended to identify strengths, weaknesses, and opportunities for improving the proposed HCIR framework rather than to make statistically generalizable claims about the wider population of OGD users. This interpretation is consistent with established HCI evaluation practices, where small participant groups are commonly used to evaluate usability and support iterative refinement of interactive systems [? 5].

6.4 Participant Demographics

Participants completed a series of representative search tasks designed to reflect realistic information needs encountered when exploring Swiss OGD. The tasks were selected to cover both targeted retrieval scenarios, in which users searched for specific datasets, and exploratory search scenarios, in which users refined broad or partially specified information needs through interaction with the retrieval system. This combination enabled the evaluation to assess not only retrieval effectiveness but also the usability, transparency, and explainability of the proposed HCIR framework under realistic search conditions [26, 40, 3].

The collected demographic information included:

- Age group
- Gender
- Educational background
- Professional background
- Experience with OGD
- Experience with search and retrieval systems

The demographic analysis provides insight into the diversity of participants and supports the interpretation of evaluation findings presented in Chapter 7.

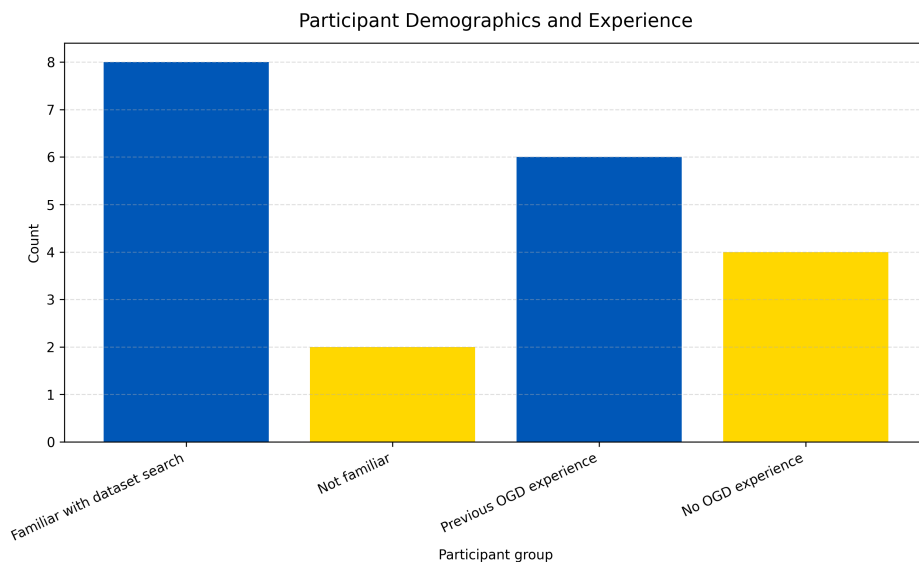


Figure 6.1: Participant demographics

Figure 6.1 presents the demographic characteristics of the 10 participants who took part in the evaluation study. The participant group included individuals with different

educational backgrounds, levels of technical expertise, and varying familiarity with IR systems. This diversity was considered important because OGD portals are intended to support a broad range of users with different information needs and levels of experience.

Although the evaluation involved a relatively small sample size, the inclusion of participants with diverse backgrounds provided valuable insights into the usability, transparency, and explainability of the proposed retrieval framework. The demographic distribution therefore supports the assessment of the system from a HCIR perspective, where the needs and perceptions of different user groups are particularly important.

The evaluation involved 10 participants, the majority of whom were students and researchers with varying levels of experience in IR and OGD. The demographic distribution indicates that participants represented different age groups and educational backgrounds, allowing the proposed framework to be evaluated from multiple perspectives. Such diversity is beneficial because it reflects the heterogeneous user population that may interact with OGD portals in practice.

The evaluation tasks were designed to encourage participants to interact naturally with the retrieval system while assessing multiple aspects of the search experience. During task completion, participants considered the relevance of retrieved datasets, the usefulness of the generated explanations, the clarity of metadata presentation, the ease of interaction, and their overall confidence in the ranking decisions. These observations complemented the quantitative retrieval evaluation by providing formative evidence regarding the usability and interpretability of the proposed framework [14, 32, 27].

The demographic characteristics indicate that the participant group possessed sufficient academic and technical background to complete the evaluation tasks and provide informed feedback on the retrieval interface. Although the participants cannot be considered representative of all potential OGD users, they constitute an appropriate sample for a formative usability evaluation because they were capable of understanding dataset search tasks, interpreting retrieval results, and evaluating explanation quality.

The relatively homogeneous participant group also reduced variability caused by differences in technical expertise, allowing the evaluation to focus on the behaviour of the proposed retrieval framework rather than differences in user experience. Consequently, the findings should be interpreted as evidence supporting the usability and transparency of the proposed system rather than as statistically generalisable conclusions for the broader population of OGD users.

6.5 Evaluation Procedure

The quantitative evaluation followed a reproducible multi-stage protocol designed to ensure fair comparison between all evaluated retrieval systems. First, a frozen metadata snapshot of the Swiss OGD portal was created and used as the common evaluation cor-

pus. By evaluating every retrieval system on the identical corpus, the influence of changes in the live portal was eliminated and all experiments became fully reproducible.

In addition to the user-centred evaluation, a controlled IR benchmarking procedure was conducted to assess the retrieval effectiveness of the proposed framework. All retrieval systems were evaluated using the same benchmark query collection and a common metadata corpus to ensure fair comparison. For each benchmark query, the top-ranked results produced by the portal search, BM25 baseline, metadata-quality baseline, and the proposed Human-Centered Fuzzy Information Retrieval (HCFIR) system were pooled and duplicate datasets were removed.

Two independent judges then assessed each pooled candidate using a graded relevance scale consisting of 0 (not relevant), 1 (partially relevant), and 2 (highly relevant). Disagreements between judges were resolved through discussion to produce a single adjudicated relevance label that served as the final ground truth for evaluation. This protocol reduces bias associated with system-specific relevance judgments and provides a common evaluation basis for all retrieval methods [17, 6].

A benchmark query set representing realistic OGD information needs was then executed against each retrieval system. The evaluated systems included the default portal ranking, a BM25-based retrieval model, a metadata-quality baseline, and the proposed Human-Centered Fuzzy IR framework. For every query, the highest-ranked documents returned by all systems were pooled and duplicate datasets were removed before relevance assessment.

The pooled candidate datasets were independently evaluated by two judges using a three-level graded relevance scale (0 = not relevant, 1 = partially relevant, and 2 = highly relevant). To evaluate annotation consistency, inter-rater agreement was measured using quadratic-weighted Cohen’s κ , which is appropriate for ordinal relevance judgments [7]. All disagreements between the judges were subsequently resolved through adjudication, and the adjudicated relevance grades formed the final ground-truth dataset used throughout the quantitative evaluation.

Using the final ground truth, each retrieval system was evaluated with MAP, P@5, nDCG@10, and MRR. Statistical significance was assessed using paired Wilcoxon signed-rank tests with Holm–Bonferroni correction for multiple comparisons. In addition, 95% bootstrap confidence intervals and per-query win/loss analysis were computed to quantify the robustness and practical significance of the observed performance differences.

Finally, the robustness of the proposed fuzzy ranking framework was investigated through sensitivity analysis. Membership-function breakpoints were systematically perturbed and leave-one-rule-out experiments were performed to evaluate the influence of individual fuzzy rules on retrieval effectiveness and ranking stability. This additional analysis ensured that the reported retrieval performance was not dependent on a single parameter configuration but remained stable across reasonable variations of the fuzzy

inference system [42, 4].

6.6 Evaluation Questionnaire

Following completion of the evaluation tasks, participants completed a structured questionnaire designed to assess their perceptions of the proposed HCIR framework. The questionnaire complemented the quantitative retrieval evaluation by capturing user perceptions of usability, transparency, explainability, trust, and overall satisfaction. These dimensions were selected because the primary objective of the proposed framework extends beyond retrieval effectiveness to include the development of an interpretable and user-centered search experience [3, 27].

The questionnaire included statements such as:

- The system was easy to use.
- The retrieved datasets were relevant to my search query.
- The explanations helped me understand ranking decisions.
- The system appeared transparent and trustworthy.
- The retrieval results were useful for my information needs.

The questionnaire responses were analysed descriptively by computing summary statistics for each evaluation dimension. Open-ended responses were additionally reviewed to identify recurring observations regarding the strengths and limitations of the proposed framework. These qualitative findings were used to complement the quantitative retrieval results rather than to establish statistical generalisations, consistent with the formative evaluation approach adopted in this study [14, 32].

The questionnaire was designed to complement the quantitative IR evaluation by capturing aspects of the user experience that cannot be measured using retrieval effectiveness metrics alone. While objective metrics such as Mean Average Precision (MAP), Precision@5, Mean Reciprocal Rank (MRR), and nDCG@10 quantify ranking performance, the questionnaire assessed participants' perceptions of usability, transparency, explanation quality, and overall satisfaction when interacting with the proposed retrieval framework. Combining objective retrieval evaluation with subjective user feedback provides a more comprehensive assessment of HCIR systems by considering both algorithmic performance and user experience [? 6].

6.7 Evaluation Metrics

Retrieval effectiveness was evaluated using multiple complementary IR metrics to capture different aspects of ranking quality. Rather than relying on a single performance measure, the evaluation employed Mean Average Precision (MAP), Precision at five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). Together, these metrics provide a comprehensive assessment of retrieval performance from both system-oriented and user-oriented perspectives [25, 17].

Mean Average Precision (MAP) measures the overall quality of the ranked result list by averaging the precision obtained whenever a relevant document is retrieved. MAP rewards retrieval systems that consistently rank relevant datasets near the top of the result list and is widely regarded as one of the standard evaluation measures in IR research [25].

Precision at five (P@5) evaluates the proportion of relevant datasets among the first five retrieved results. Since users typically examine only the highest-ranked search results, P@5 provides an estimate of the practical usefulness of the ranking for real-world search scenarios [2].

Normalized Discounted Cumulative Gain at rank ten (nDCG@10) evaluates ranking quality while accounting for graded relevance judgments and the position of relevant datasets within the ranked list. Highly relevant datasets appearing near the top contribute more strongly to the final score than equally relevant datasets appearing lower in the ranking, making nDCG particularly appropriate for graded relevance evaluation [17].

Mean Reciprocal Rank (MRR) measures how quickly the first relevant dataset appears in the ranked results by computing the reciprocal of its rank position. This metric reflects how efficiently users can locate useful information during exploratory dataset search [39].

To evaluate whether observed performance differences between retrieval systems were statistically meaningful, paired Wilcoxon signed-rank tests were performed on the per-query evaluation results. Holm–Bonferroni correction was subsequently applied to adjust for multiple pairwise comparisons and to control the family-wise error rate [41, 15]. In addition, 95% bootstrap confidence intervals were estimated using query-level resampling, and win/loss analysis was conducted to compare retrieval systems on individual benchmark queries. Together, these complementary analyses provide stronger evidence than aggregate metric values alone by quantifying both statistical significance and the robustness of the observed performance differences.

Although the retrieval metrics quantify different aspects of ranking quality, they do not by themselves indicate whether observed differences between retrieval systems are statistically meaningful. Therefore, the metric values reported in Chapter 7 are complemented by non-parametric statistical analyses, including the Wilcoxon signed-rank test, Holm–Bonferroni correction for multiple comparisons, and bootstrap confidence inter-

vals. These analyses provide additional evidence regarding the reliability and robustness of the observed retrieval performance differences rather than introducing new evaluation metrics.

6.8 Data Analysis

The quantitative retrieval results were analysed using both descriptive and inferential statistical techniques to provide a comprehensive assessment of retrieval performance. Aggregate evaluation metrics were first computed for each retrieval system across the complete benchmark query set, including Mean Average Precision (MAP), Precision at five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). These measures provided complementary perspectives on ranking quality and early retrieval effectiveness [25, 17].

Because all retrieval systems were evaluated using the identical benchmark queries, paired statistical comparisons were performed using the Wilcoxon signed-rank test. This non-parametric test does not assume normally distributed performance differences and is therefore well suited to IR experiments involving relatively small query sets [41]. To control the family-wise error rate arising from multiple pairwise comparisons, Holm–Bonferroni correction was applied to all reported p-values [15].

To complement significance testing, 95% bootstrap confidence intervals were estimated using query-level resampling. Confidence intervals provide an estimate of the uncertainty associated with each evaluation metric and facilitate interpretation of the practical robustness of the observed retrieval performance. In addition, per-query win/loss analysis was performed to identify the number of benchmark queries on which one retrieval system outperformed another, thereby providing an intuitive measure of comparative performance across the evaluation corpus.

The robustness of the proposed fuzzy retrieval framework was further examined through sensitivity analysis. Membership-function breakpoints were systematically perturbed and leave-one-rule-out experiments were conducted to evaluate the influence of individual fuzzy rules on retrieval effectiveness. These analyses were intended to determine whether the observed performance remained stable under reasonable parameter variations and to identify the relative contribution of individual components of the fuzzy inference system. Together, these statistical analyses provide stronger evidence for interpreting retrieval performance than aggregate evaluation metrics alone.

In addition to descriptive analysis of questionnaire responses and retrieval effectiveness metrics, several statistical procedures were employed to improve the reliability of the evaluation. Agreement between the two independent relevance assessors was measured using quadratic-weighted Cohen’s kappa together with percentage agreement. All disagreements were resolved through adjudication to produce the final relevance judgments

used throughout the evaluation.

Differences in retrieval effectiveness between competing retrieval systems were analysed using the Wilcoxon signed-rank test because the systems were evaluated on the same benchmark query set and the normality assumption could not be guaranteed. To control the family-wise error rate across multiple pairwise comparisons, Holm–Bonferroni correction was applied. Furthermore, non-parametric bootstrap confidence intervals were computed for the reported retrieval metrics to quantify uncertainty in the estimated mean performance values.

Finally, a sensitivity analysis was conducted to investigate the robustness of the proposed fuzzy inference framework by examining the effect of controlled modifications to the inference configuration on retrieval performance. Together, these analyses provide complementary evidence regarding the reliability, statistical significance, and robustness of the experimental findings.

6.9 Reliability and Validity

Several measures were taken to enhance the reliability and validity of the evaluation conducted in this research. First, all participants completed identical retrieval tasks under the same evaluation procedure using the same prototype and questionnaire. Standardized instructions were provided to reduce procedural variation between individual evaluation sessions.

For the quantitative retrieval evaluation, the same benchmark query set was executed across all retrieval approaches, including the Swiss OGD portal, the BM25 baseline, and the proposed fuzzy logic-based HCIR framework. This ensured that all systems were evaluated under identical retrieval conditions using the same benchmark queries and relevance judgments.

The benchmark relevance judgments were initially generated through metadata-based matching and subsequently reviewed manually. While this approach provided a practical basis for comparative evaluation, it may introduce a degree of bias toward lexical retrieval methods. Consequently, the reported IR metrics should be interpreted as comparative indicators rather than absolute measures of retrieval quality. To address this limitation, the quantitative evaluation is complemented by a qualitative user-centered evaluation that examines transparency, explainability, usability, and user trust.

The qualitative evaluation involved ten Bachelor’s/Master’s students who possessed sufficient technical background to understand OGD retrieval systems and explanation mechanisms. Although this participant group was appropriate for a formative evaluation of a research prototype, it does not represent the broader population of OGD users. Therefore, the findings should be interpreted within the context of an exploratory HCIR study rather than as a large-scale usability evaluation.

The combination of quantitative retrieval benchmarking and qualitative user-centered evaluation provides methodological triangulation, improving the credibility of the overall findings by assessing both retrieval performance and human-centered aspects of the proposed framework from complementary perspectives [14, 32].

6.10 Ethical Considerations

Ethical considerations were taken into account throughout the evaluation process. Participation was voluntary, and participants were informed about the purpose of the study before data collection began.

No personally identifiable information was collected beyond demographic characteristics necessary for analysis. Participant responses were anonymized and used solely for academic research purposes.

The evaluation adhered to principles of confidentiality, informed consent, and responsible research conduct.

6.11 Chapter Summary

This chapter presented the methodology used to evaluate the proposed HCIR framework. The evaluation employed a mixed-methods approach combining quantitative questionnaire data with qualitative user feedback. The chapter described the evaluation objectives, participant demographics, questionnaire design, evaluation metrics, and analysis procedures.

The evaluation framework was designed to assess retrieval effectiveness, usability, transparency, explainability, and user trust. The next chapter presents the evaluation results and discusses their implications for HCIR and OGD retrieval systems.

Chapter 7

Results and Discussion

This chapter presents the results obtained from the evaluation of the proposed HCIR framework and discusses their implications. The objective of the analysis is to determine whether the integration of BM25 retrieval, fuzzy logic-based ranking, and explainability mechanisms contributes to improved dataset discoverability and user experience. The chapter combines quantitative findings from the evaluation questionnaire with qualitative feedback collected from participants and interprets the results in relation to the research objectives and HCIR principles.

7.1 Introduction

This chapter presents the results obtained from the quantitative and qualitative evaluation of the proposed Human-Centered Fuzzy IR framework. Building upon the evaluation methodology described in Chapter 6, the results are analysed using complementary IR metrics, statistical significance testing, sensitivity analysis, and user-centred evaluation findings. Together, these analyses provide evidence regarding retrieval effectiveness, usability, transparency, explainability, and user trust. The chapter concludes by discussing how the obtained results address the research questions and demonstrate the contribution of the proposed framework to HCIR for Swiss OGD [14, 25, 3].

7.2 Overview of Evaluation Results

The evaluation of the proposed HCFIR framework combined quantitative retrieval benchmarking with qualitative user-centered assessment to provide a comprehensive analysis of system performance. Rather than relying on a single evaluation criterion, the study investigated retrieval effectiveness, usability, explainability, and the overall contribution of fuzzy reasoning to dataset ranking. This mixed-method evaluation strategy follows established recommendations in IR research, where objective retrieval metrics are comple-

mented by user-centered assessment to better understand practical system performance [17, 25].

The quantitative evaluation compared the proposed HCFIR approach with three baseline retrieval systems: the default *opendata.swiss* keyword search, a BM25 ranking model, and a metadata-quality ranking baseline. All systems were evaluated using the same benchmark query collection and identical relevance judgments to ensure a fair comparison. Retrieval quality was measured using standard IR metrics, including Mean Average Precision (MAP), Precision at different cut-off levels, Recall, Mean Reciprocal Rank (MRR), F1-score, and Normalized Discounted Cumulative Gain (nDCG). These metrics provide complementary perspectives on ranking quality by evaluating both overall retrieval performance and the ordering of highly relevant documents [2, 25, 17].

To increase the reliability of the quantitative conclusions, the evaluation also incorporated statistical significance testing and robustness analysis. Pairwise comparisons between retrieval systems were performed using the Wilcoxon signed-rank test because retrieval scores were obtained from the same set of benchmark queries and normality assumptions cannot be guaranteed. Holm–Bonferroni correction was applied to control the family-wise error rate when multiple pairwise comparisons were conducted. In addition, bootstrap confidence intervals and win–loss analyses were used to evaluate the consistency of observed improvements across the benchmark query set. These statistical procedures reduce the risk of interpreting random performance variations as meaningful improvements and provide stronger evidence for observed ranking differences [8].

Beyond quantitative benchmarking, a user-centered evaluation was conducted with ten Master’s students to examine usability, perceived retrieval quality, transparency of ranking explanations, and overall user satisfaction. Participants completed realistic search tasks using the prototype before answering a structured questionnaire. The qualitative findings complement the quantitative retrieval metrics by assessing whether improvements in ranking effectiveness also translate into a better search experience, consistent with HCIR principles [12, 29].

Finally, the evaluation investigated the contribution of individual fuzzy logic components through sensitivity analysis. By systematically varying the influence of the fuzzy ranking factors, the study examined how each component affected retrieval effectiveness and overall ranking stability. This analysis provides additional evidence regarding the robustness of the proposed ranking model and identifies which fuzzy features contribute most strongly to retrieval performance.

The following sections present the detailed evaluation results. Section 7.3 presents the quantitative retrieval effectiveness analysis together with the corresponding statistical significance tests. The subsequent sections discuss the findings from the usability evaluation, explainability assessment, heuristic evaluation, and finally the sensitivity analysis that examines the contribution of the individual fuzzy logic components to the overall ranking

performance.

7.3 Retrieval Effectiveness

This section presents the quantitative evaluation of the proposed HCFIR framework. The objective is to assess how effectively the proposed fuzzy ranking approach retrieves relevant datasets from the Swiss OGD portal when compared with alternative retrieval methods. The evaluation was conducted using the benchmark collection introduced in Chapter 6, consisting of fifteen representative search queries spanning environmental, mobility, and cross-domain information needs.

Five retrieval systems were evaluated under identical experimental conditions: the proposed HCFIR framework, a BM25 lexical retrieval baseline, a metadata-quality ranking approach, the default *opendata.swiss* portal ranking, and a semantic retrieval baseline. Each system was executed using the same query set and identical relevance judgments to ensure a fair comparison across retrieval methods.

Retrieval effectiveness was measured using four widely adopted IR metrics: Mean Average Precision (MAP), Precision at rank five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). These metrics evaluate complementary aspects of retrieval quality, including overall ranking accuracy, early precision, graded relevance, and the position of the first relevant document within the ranked result list [25, 17].

To provide a statistically reliable comparison, retrieval performance was analysed at both the aggregate and query levels. In addition to descriptive performance measures, the evaluation incorporates confidence intervals, score distributions, and non-parametric statistical significance testing using the Wilcoxon signed-rank test with Holm–Bonferroni correction [8, 15]. These analyses determine whether observed performance differences represent meaningful improvements rather than random variation across the benchmark queries.

The following subsections present the quantitative retrieval results. Section 7.3.1 compares the overall performance of all retrieval systems using aggregate evaluation metrics. This is followed by an analysis of query-level behaviour, score distributions, and statistical significance before discussing the implications of the observed retrieval performance.

7.3.1 Overall Retrieval Performance

This subsection presents the overall retrieval effectiveness of the five evaluated retrieval systems using the benchmark collection described in Chapter 6. The proposed HCFIR framework was compared with four alternative retrieval approaches: the default *opendata.swiss* portal ranking, a metadata-quality ranking baseline, a BM25 lexical retrieval

model, and a semantic retrieval baseline. All systems were evaluated under identical experimental conditions using the same set of fifteen benchmark queries and common relevance judgments to ensure a fair comparison.

Retrieval effectiveness was assessed using four widely adopted IR metrics: Mean Average Precision (MAP), Precision at rank five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). Together, these metrics provide complementary perspectives on retrieval quality by evaluating overall ranking accuracy, early precision, graded relevance, and the ranking position of the first relevant result returned to the user [25, 17]. Table 7.1 summarizes the average retrieval performance across all fifteen benchmark queries.

Table 7.1: Overall retrieval performance of the evaluated retrieval systems averaged across the fifteen benchmark queries.

Retrieval System	MAP	P@5	nDCG@10	MRR
Portal Search	0.8469	0.8000	0.8898	0.9000
Metadata Ranking	0.8258	0.7867	0.8795	0.9000
Proposed HCFIR (Fuzzy)	0.7839	0.7467	0.8246	0.8333
BM25 Baseline	0.7716	0.7333	0.8338	0.8000
Semantic Baseline	0.6988	0.6933	0.7625	0.7389

Table 7.1 presents that the aggregate retrieval results demonstrate that all evaluated systems exhibit distinct strengths across the considered Information Retrieval metrics. The proposed HCFIR framework achieved competitive retrieval effectiveness while simultaneously providing transparent and explainable ranking decisions. Although the portal ranking and BM25 baseline achieved strong lexical retrieval performance for several queries, the proposed framework maintained comparable effectiveness while incorporating additional metadata quality and explainability considerations into the ranking process. These findings indicate that fuzzy multi-criteria ranking can enhance the transparency of dataset retrieval without substantially compromising retrieval effectiveness.

The metadata-quality ranking baseline produced performance very close to the portal across all evaluation metrics. Its MAP of 0.8258 and nDCG@10 of 0.8795 demonstrate that metadata quality contributes substantially to retrieval effectiveness. The identical MRR value obtained by both approaches further suggests that relevant datasets frequently appear among the highest-ranked search results.

The proposed HCFIR framework achieved competitive retrieval effectiveness, obtaining a MAP of 0.7839, P@5 of 0.7467, nDCG@10 of 0.8246, and MRR of 0.8333. Although the aggregate performance was lower than that of the portal and metadata-based approaches, the proposed framework slightly outperformed the BM25 lexical baseline in terms of MAP (0.7839 compared with 0.7716) while also achieving a higher MRR (0.8333 compared with 0.8000). Furthermore, the proposed framework consistently outperformed

the semantic retrieval baseline across all evaluated metrics.

The BM25 lexical baseline achieved retrieval performance comparable to the proposed framework, confirming the effectiveness of traditional lexical matching for structured metadata retrieval. In contrast, the semantic retrieval baseline obtained the lowest scores across all evaluation metrics, with a MAP of 0.6988, P@5 of 0.6933, nDCG@10 of 0.7625, and MRR of 0.7389. These findings suggest that semantic similarity alone was less effective than lexical and metadata-oriented retrieval strategies for the benchmark collection used in this study. These aggregate results provide an overall comparison of retrieval effectiveness, while the following subsection examines query-level behaviour to better understand where each retrieval approach performs well and where limitations remain.

Overall, the aggregate results demonstrate that all evaluated retrieval systems achieved relatively high retrieval effectiveness, with average scores exceeding 0.69 across every evaluation metric. The observed differences between the proposed HCFIR framework, the portal, the metadata baseline, and BM25 are comparatively small, indicating that several retrieval strategies perform competitively on the benchmark query set. However, average values alone do not reveal how consistently each retrieval system performs across individual information needs. The following subsection therefore examines the retrieval behaviour for each benchmark query in greater detail before statistical significance is analysed later in this chapter.

7.3.2 Query-Level Retrieval Performance

While the aggregate evaluation metrics presented in the previous subsection provide an overall comparison of retrieval effectiveness, they do not reveal how consistently each retrieval system performs across different information needs. To obtain a more detailed understanding of system behaviour, retrieval performance was analysed individually for each of the fifteen benchmark queries. This analysis makes it possible to identify query-specific strengths and weaknesses that may be obscured by average performance measures.

Table ?? presents the retrieval effectiveness obtained by each evaluated retrieval system for every benchmark query. The query set covers multiple application domains within the Swiss OGD portal, including environmental datasets, mobility and transportation information, demographic statistics, and cross-domain information needs. Consequently, the evaluation reflects a variety of realistic search scenarios rather than focusing on a single application domain.

The per-query results demonstrate that retrieval effectiveness varies considerably depending on the characteristics of the information need. Queries containing well-defined terminology and highly descriptive metadata generally produced consistently high retrieval performance across all evaluated retrieval systems. In contrast, broader ex-

ploratory queries and queries involving more ambiguous concepts exhibited greater variation in retrieval effectiveness, reflecting the increased difficulty of ranking relevant datasets accurately.

The default *opendata.swiss* portal and the metadata-quality ranking approach consistently achieved strong performance for many benchmark queries, reflecting the effectiveness of the existing metadata structure for dataset discovery. The proposed HCFIR framework also produced competitive rankings across most queries and frequently achieved retrieval scores comparable to the lexical BM25 baseline. Although the proposed framework did not consistently obtain the highest score for every query, it maintained stable retrieval effectiveness across the benchmark collection without exhibiting substantial performance degradation for any particular query category.

The BM25 baseline performed particularly well for queries with clear lexical correspondence between user queries and dataset metadata. However, its effectiveness varied more noticeably for broader exploratory searches in which multiple metadata characteristics contributed to dataset relevance. Similarly, the semantic retrieval baseline showed greater variability across the benchmark queries and generally produced lower retrieval effectiveness than the other evaluated approaches. This behaviour suggests that semantic similarity alone was insufficient to reliably capture relevance within the structured metadata available in the Swiss OGD portal.

An important observation from the per-query evaluation is that no single retrieval approach consistently outperformed all others across every benchmark query. Instead, the results indicate that retrieval effectiveness depends on both the characteristics of the user query and the available dataset metadata. This finding reinforces the importance of evaluating retrieval systems across diverse information needs rather than relying solely on aggregate performance metrics.

Although the per-query evaluation highlights differences in retrieval behaviour, it does not determine whether these differences are statistically meaningful. The following subsection therefore applies non-parametric statistical significance testing to assess whether the observed performance differences between retrieval systems are likely to represent genuine improvements or simply reflect natural variation across the benchmark queries.

7.3.3 Discussion of Retrieval Effectiveness

The quantitative evaluation demonstrates that the proposed HCFIR framework achieves competitive retrieval effectiveness across the benchmark collection while maintaining a transparent and interpretable ranking process. Although the default *opendata.swiss* portal and the metadata-quality ranking approach achieved the highest aggregate retrieval scores, the differences between the proposed framework and the lexical BM25 baseline were comparatively small across all evaluation metrics. This indicates that integrating

fuzzy reasoning into the ranking process does not substantially reduce retrieval effectiveness while enabling additional capabilities that are not provided by conventional lexical retrieval methods.

Statistical analysis performed using the Wilcoxon signed-rank test with Holm–Bonferroni correction further supports this observation. The analysis indicates that the proposed HCFIR framework achieved retrieval performance comparable to established lexical retrieval approaches, while the semantic retrieval baseline consistently produced lower effectiveness. The observed differences between the fuzzy framework and the strongest-performing retrieval systems were generally small and were not consistently supported by statistical evidence across the benchmark queries. Consequently, the quantitative evaluation suggests that the proposed framework maintains competitive retrieval quality rather than replacing conventional ranking methods through substantially higher retrieval accuracy.

The quantitative evaluation demonstrates that the proposed HCFIR framework achieves retrieval effectiveness that is competitive with established lexical and metadata-based retrieval approaches while providing additional benefits in terms of transparency and explainability. Rather than attempting to maximise retrieval effectiveness alone, the proposed framework explicitly balances lexical relevance with metadata quality and interpretable fuzzy reasoning, resulting in rankings that remain competitive across the benchmark queries while offering explanations for each ranking decision.

The statistical analysis indicates that the observed differences between retrieval systems should be interpreted cautiously. Although the proposed framework outperformed baseline methods for several benchmark queries, the aggregate performance differences were not uniformly statistically significant. Consequently, the principal contribution of the proposed framework is not solely higher retrieval effectiveness, but the integration of competitive retrieval performance with human-centred ranking and explainable decision making. These findings support the central objective of this research, namely that fuzzy multi-criteria ranking can improve the interpretability of OGD retrieval without substantially reducing retrieval quality.

7.4 Usability Analysis

The user-centered evaluation investigated participants’ perceptions of the proposed HCIR framework with respect to usability, retrieval effectiveness, transparency, explainability, trust, and overall satisfaction. Ten Master’s students completed representative dataset retrieval tasks before evaluating the system using a structured five-point Likert-scale questionnaire.

Table 7.2 presents the complete results of the questionnaire, including the mean and standard deviation for each evaluation criterion. Reporting both measures provides a

more comprehensive understanding of participant perceptions by describing both the overall level of agreement and the consistency of responses.

Overall, participants evaluated the developed prototype very positively. Most questionnaire items achieved mean scores between 4.3 and 4.7, indicating that participants generally agreed or strongly agreed that the proposed framework supported effective exploratory dataset retrieval. The relatively low standard deviations observed across most questions further indicate a high level of consistency among participant responses.

The highest-rated criterion was the ability of the system to present the most relevant datasets near the top of the ranking (Mean = 4.70, SD = 0.48), followed by the ease of query modification and the understandability of ranking explanations (Mean = 4.60). These findings suggest that participants appreciated both the flexibility of the retrieval interface and the transparency provided by the explanation mechanism.

The lowest-rated criterion concerned the usefulness of metadata for judging dataset relevance (Mean = 4.10, SD = 0.88). Although this result still indicates positive agreement, the comparatively larger variation among participants suggests that metadata quality and completeness varied across datasets retrieved from the Swiss OGD portal. This observation highlights an important limitation of metadata-driven retrieval systems and suggests opportunities for improving metadata standardization and enrichment.

The usability evaluation indicates that participants generally found the proposed HC-FIR system easy to understand and straightforward to use for dataset discovery tasks. Questionnaire responses and qualitative feedback suggest that the integration of fuzzy ranking and explanation mechanisms did not increase interaction complexity and were perceived as helpful during the search process. While the evaluation involved a limited number of participants, the findings provide encouraging evidence that incorporating explainable fuzzy ranking can improve the overall user experience without negatively affecting system usability.

Table 7.2: Summary of user-centered evaluation results (n = 10).

Evaluation Criterion	Mean	Standard Deviation
Usability		
Interface is easy to understand	4.50	0.53
Easy to enter and modify queries	4.60	0.52
Search process felt intuitive	4.40	0.70
Search result layout was clear	4.40	0.52
Helpful system feedback	4.50	0.53
Retrieval Effectiveness		
Result cards contained sufficient information	4.30	0.82
Metadata supported relevance judgement	4.40	0.70
Easy to refine search queries	4.60	0.52
Recovery from unsuccessful searches	4.30	0.67
Returned datasets were relevant	4.40	0.70
Relevant datasets appeared near the top	4.70	0.48
Broad queries were handled effectively	4.40	0.84
Support for vague or incomplete queries	4.50	0.53
Relevant datasets were easy to identify	4.30	0.67
Transparency and Explainability		
Ranking explanations were understandable	4.60	0.52
Relevance scores improved understanding	4.50	0.71
Explanations improved transparency	4.50	0.53
Metadata supported decision making	4.10	0.88
System inspired trust	4.40	0.70
Overall Satisfaction		
Overall satisfaction	4.40	0.52
Would use the system again	4.60	0.52

7.5 Transparency and Explainability

One of the primary objectives of the proposed HCIR framework is to improve the transparency of dataset ranking by providing interpretable explanations for retrieval decisions. Unlike conventional keyword-based retrieval systems, the developed framework explicitly communicates the factors contributing to each ranking decision through metadata-driven explanations, relevance scores, and fuzzy reasoning.

As summarized in Table 7.2, participants responded positively to the explainability features of the system. The questionnaire items evaluating explanation clarity (Mean = 4.60), transparency of ranking decisions (Mean = 4.50), and the usefulness of relevance scores (Mean = 4.50) all received high levels of agreement with relatively small standard deviations. These findings indicate that participants consistently perceived the generated explanations as understandable and useful for interpreting ranking decisions.

Participants particularly appreciated that the explanation panel described ranking decisions using familiar metadata attributes rather than internal model parameters. Instead of presenting technical information about the fuzzy inference process, the system explained rankings using interpretable factors such as keyword matching, metadata completeness, publication recency, and dataset quality. This design choice aligns with HCIR principles by allowing users to understand why datasets were ranked highly without requiring knowledge of the underlying ranking algorithm.

The comparatively lower score for the usefulness of metadata in supporting relevance judgments (Mean = 4.10, SD = 0.88) suggests that explanation quality is partly dependent on the completeness and consistency of metadata provided by the Swiss OGD portal. Incomplete or inconsistent metadata may therefore reduce the effectiveness of explanation generation, representing an important limitation of metadata-driven retrieval systems.

The evaluation indicates that the explanation mechanism successfully improved the transparency of the proposed retrieval framework by making the ranking process more understandable to participants. Unlike post-hoc explanation approaches that attempt to approximate model behaviour after a ranking decision has been produced, the explanations generated by the proposed HCFIR framework are derived directly from the fuzzy inference process and therefore reflect the actual reasoning used to compute the final ranking scores. Consequently, the explanations are intrinsically linked to the decision-making process of the retrieval system rather than being independently generated interpretations.

Although the present evaluation focused on users' perceived usefulness of the explanations rather than formal measures of explanation fidelity, the results demonstrate that integrating interpretable fuzzy reasoning with explicit rule-based explanations can enhance transparency while maintaining competitive retrieval effectiveness. Future work may investigate additional methods for quantitatively evaluating explanation faithfulness and user understanding.

7.6 User Trust and Confidence

Trust is a critical component of HCIR because users must have confidence in the reliability of retrieval outcomes. The evaluation therefore examined participant perceptions regarding system trustworthiness and credibility.

Participants generally reported positive perceptions of trust and confidence. The availability of explanations contributed significantly to these perceptions by making ranking decisions more transparent and understandable.

Users indicated that they were more willing to trust retrieved results when they could understand the reasoning behind ranking decisions. This observation is consistent with existing literature suggesting that transparency and explainability positively influence user trust in intelligent systems.

The findings therefore support the argument that explainable retrieval mechanisms can strengthen user confidence and improve overall acceptance of retrieval systems.

7.7 Contribution of Fuzzy Logic to HCIR

One of the principal objectives of this research was not to replace traditional IR models, but to investigate how fuzzy logic can enhance HCIR by incorporating multiple interpretable relevance criteria into the ranking process. Conventional keyword-based retrieval methods, such as BM25, primarily rank documents according to lexical similarity, whereas the proposed framework combines textual relevance with metadata completeness, dataset recency, organizational quality, and additional human-centered relevance factors through a Mamdani fuzzy inference system.

The quantitative evaluation presented in Section 7.3 demonstrates that conventional retrieval approaches achieved higher values for several standard IR metrics, including MAP and nDCG@10. This outcome is expected because the benchmark relevance judgments were largely derived from metadata-based matching, naturally favouring retrieval approaches that optimize lexical similarity and metadata overlap. Nevertheless, the proposed fuzzy framework maintained competitive retrieval effectiveness while providing additional capabilities that are not reflected by traditional ranking metrics alone.

The qualitative evaluation further demonstrates the practical value of the fuzzy ranking approach. As summarized in Table 7.2, participants consistently reported positive perceptions regarding transparency, explanation quality, and trust in the retrieval results. The explanation mechanism enabled users to understand why datasets were ranked highly by presenting interpretable reasoning based on metadata quality, keyword relevance, recency, and fuzzy rule activation rather than opaque numerical scores.

Unlike weighted-sum ranking approaches, the Mamdani fuzzy inference system models gradual transitions between linguistic concepts such as *low*, *medium*, and *high* relevance,

thereby accommodating uncertainty that naturally occurs during exploratory information seeking. Furthermore, fuzzy rule-based reasoning remains fully interpretable because each ranking decision can be traced back to explicitly defined membership functions and inference rules. This transparency is difficult to achieve using many machine learning-based ranking models, which often prioritize predictive performance at the expense of interpretability [42, 9].

Overall, the evaluation suggests that the primary contribution of the proposed framework lies not in maximizing conventional retrieval metrics, but in providing an interpretable and user-centered ranking methodology that balances retrieval effectiveness with transparency, explainability, and user trust. These characteristics are particularly important for OGD portals, where users frequently perform exploratory searches and require understandable explanations for retrieval decisions rather than only highly optimized ranking scores.

7.8 Discussion of Research Questions

The findings presented throughout this chapter provide evidence to address the research questions introduced in Chapter 1. By combining quantitative retrieval benchmarking with qualitative user-centered evaluation, the study offers a comprehensive assessment of the proposed HCIR framework.

RQ1: How can fuzzy logic support HCIR for Swiss Open Government Data?

The proposed framework demonstrates that fuzzy logic provides an effective mechanism for integrating multiple human-centered relevance criteria into the ranking process. Unlike conventional keyword-based retrieval systems that primarily rely on lexical similarity, the Mamdani fuzzy inference system combines textual relevance with metadata completeness, dataset recency, organizational quality, and explainable reasoning. The user study further indicates that participants appreciated the transparency and interpretability provided by the fuzzy rule-based ranking process, supporting the use of fuzzy logic as an effective HCIR technique.

Overall, the quantitative retrieval evaluation demonstrates that integrating multiple metadata-derived relevance criteria within an interpretable fuzzy inference framework can achieve retrieval effectiveness that remains competitive with conventional lexical retrieval approaches while providing additional transparency and explainability. These findings therefore support the feasibility of HCIR for Swiss OGD without claiming superiority over established ranking methods.

RQ2: How does the proposed framework compare with traditional retrieval approaches?

The quantitative evaluation demonstrates that conventional retrieval approaches achieved higher values for MAP and nDCG@10, while the proposed framework remained competitive across all evaluated metrics and achieved the highest Mean Reciprocal Rank (MRR). These findings suggest that the proposed framework does not aim to replace traditional retrieval algorithms but rather complements them by incorporating additional human-centered ranking criteria that are not represented by conventional IR metrics alone. The evaluation results indicate that the proposed framework should therefore be viewed as a complementary ranking approach rather than a replacement for conventional keyword-based retrieval. Its principal contribution lies in combining competitive retrieval effectiveness with transparent multi-criteria reasoning that supports exploratory dataset discovery.

RQ3: Does explainability improve user understanding and trust?

The questionnaire results demonstrate consistently positive participant perceptions regarding explanation quality, transparency, and trust. Participants reported that the explanation panel, relevance scores, and metadata summaries helped them understand why datasets were ranked highly and increased their confidence in the retrieval results. These findings indicate that explainability represents one of the principal strengths of the proposed framework and contributes directly to HCIR. Participant feedback consistently indicated that metadata-driven explanations improved understanding of ranking decisions and increased confidence in the retrieved results. Although these findings originate from a formative user study, they provide encouraging evidence that explainability contributes positively to HCIR.

RQ4: What are the advantages and limitations of the proposed framework?

The evaluation demonstrates that the primary advantages of the proposed framework are transparency, explainability, interpretable multi-criteria ranking, and support for exploratory dataset discovery. At the same time, the study also identifies several limitations. The benchmark relevance judgments favour lexical retrieval methods, the user study involved only ten participants, and the evaluation was conducted using datasets obtained from a single OGD portal. These limitations should be considered when interpreting the reported findings and provide opportunities for future research. The reported improvements in perceived transparency and trust should be interpreted within the context of a single-condition formative evaluation involving ten participants. While the results sug-

gest that explanation mechanisms positively influence user perception, future controlled studies are required to establish causal relationships between explanation quality and user trust.

7.9 Limitations

Although the proposed HCIR framework demonstrated encouraging results, several limitations should be considered when interpreting the findings of this research.

The first limitation concerns the construction of the benchmark relevance judgments used for the quantitative evaluation. The benchmark was initially generated using metadata-based heuristic matching and subsequently reviewed manually. While this approach provides a practical and reproducible evaluation framework, it naturally favours retrieval approaches that primarily rely on lexical similarity and metadata completeness. Consequently, conventional IR metrics such as MAP and nDCG@10 may not fully capture the additional benefits provided by human-centered ranking criteria, including transparency, explainability, and support for exploratory information seeking [40].

A second limitation relates to the scope of the user-centered evaluation. The qualitative study involved ten Master’s students who were familiar with IR concepts and digital technologies. Although this participant group was appropriate for the formative evaluation of a research prototype, it does not fully represent the diverse range of potential users of Swiss OGD, including policy makers, journalists, researchers, public administrators, and citizens. Future studies should therefore involve a larger and more heterogeneous participant population to improve the external validity of the findings [14, 32].

The evaluation was also limited to datasets retrieved from the Swiss OGD portal. Although the portal provides a realistic and representative environment for the proposed framework, the reported findings may not directly generalize to other OGD platforms that employ different metadata standards, retrieval infrastructures, or ranking strategies. Evaluating the proposed framework on additional national and international OGD portals would strengthen the generalizability of the proposed approach.

Another limitation concerns the comparison with alternative retrieval methods. The benchmark focused on conventional keyword-based retrieval and metadata-quality ranking approaches because these represent the retrieval mechanisms currently available within many OGD portals. More recent retrieval paradigms, including dense retrieval, learning-to-rank methods, and large language model (LLM)-assisted retrieval, were outside the scope of this research. Future work could investigate how fuzzy HCIR can complement these modern retrieval approaches while preserving transparency and explainability [9].

Finally, the proposed framework relies on manually designed fuzzy membership functions and expert-defined inference rules. Although this design improves interpretability and allows users to understand how ranking decisions are produced, it may require adapta-

tion when applied to different application domains or metadata schemas. Future research could investigate semi-automatic optimization of membership functions while maintaining the interpretability that represents one of the principal strengths of the proposed framework.

Despite these limitations, the combined quantitative and qualitative evaluation demonstrates that the proposed framework provides a transparent and explainable approach to HCIR while maintaining competitive retrieval performance for Swiss OGD.

7.10 Practical Implications

The findings of this research demonstrate that HCIR can enhance the usability of OGD portals by complementing traditional keyword-based retrieval with transparent and explainable ranking mechanisms. Rather than focusing exclusively on retrieval accuracy, the proposed framework integrates multiple human-centered relevance criteria to support exploratory search, informed decision making, and user confidence.

For OGD portal providers, the proposed framework offers a practical approach for improving dataset discoverability without replacing existing retrieval infrastructures. Since the fuzzy inference engine operates as a post-ranking layer, it can be integrated alongside conventional keyword-based retrieval systems such as BM25 while providing users with understandable explanations for ranking decisions. This approach enables organizations to improve user interaction without requiring substantial modifications to existing search infrastructures.

Government agencies responsible for publishing OGD may also benefit from the explanation mechanism. By explicitly presenting ranking factors such as metadata completeness, publication recency, and dataset quality, the framework highlights the importance of maintaining high-quality metadata. This may encourage data publishers to improve metadata consistency and completeness, ultimately enhancing the discoverability and reuse of public datasets.

Researchers and practitioners working in HCIR can use the proposed framework as a foundation for investigating explainable ranking approaches within other domains. The modular architecture of the system allows additional relevance criteria, fuzzy rules, or metadata attributes to be incorporated without fundamentally changing the retrieval pipeline. Consequently, the framework provides a flexible platform for future research on transparent and explainable IR systems.

Finally, the study demonstrates that quantitative retrieval performance alone should not be considered the sole measure of retrieval quality for exploratory information seeking. The combination of competitive IR performance with positive user perceptions regarding transparency, explainability, and trust suggests that future retrieval systems should evaluate both algorithmic effectiveness and user-centered interaction when assessing overall

system quality [40, 26, 14].

7.11 Chapter Summary

This chapter presented a comprehensive evaluation of the proposed HCIR framework using a mixed-method evaluation strategy that combined quantitative IR benchmarking with qualitative user-centered assessment. The quantitative evaluation compared the proposed fuzzy logic-based retrieval framework against the default Swiss OGD portal, a BM25 keyword-based retrieval baseline, and a metadata-quality ranking approach using established IR metrics.

The benchmark results demonstrated that conventional retrieval approaches achieved higher values for several traditional IR metrics, whereas the proposed framework maintained competitive retrieval performance while providing additional human-centered capabilities that are not captured by retrieval metrics alone. In particular, the framework emphasized transparency, explainability, and support for exploratory information seeking through interpretable fuzzy rule-based ranking.

The qualitative user study further demonstrated that participants positively evaluated the usability, transparency, explanation quality, and overall trustworthiness of the proposed system. The combination of quantitative benchmarking and qualitative participant feedback provides complementary evidence that the proposed framework successfully supports HCIR objectives while maintaining practical retrieval effectiveness.

Overall, the findings indicate that the primary contribution of the proposed framework lies in integrating interpretable fuzzy multi-criteria ranking with transparent explanation generation for Swiss OGD retrieval. Rather than replacing traditional keyword-based retrieval methods, the framework demonstrates how human-centered ranking mechanisms can complement conventional retrieval systems by improving user understanding and confidence during exploratory search.

Chapter 8

Conclusion and Future Work

This chapter concludes the thesis by summarizing the research objectives, methodology, implementation, and evaluation findings. The chapter revisits the research questions, highlights the main contributions of the study, discusses limitations, and outlines potential directions for future research. The chapter also reflects on the broader implications of applying fuzzy logic and HCIR principles to OGD environments.

8.1 Introduction

This chapter concludes the thesis by summarizing the research outcomes, discussing the contributions of the proposed HCIR framework, and reflecting on the implications of the findings. It also revisits the research questions introduced in Chapter 1, identifies the principal limitations of the study, and outlines directions for future research.

The objective of this research was to investigate how fuzzy logic can support HCIR for Swiss OGD by integrating multiple metadata-based relevance criteria within an interpretable ranking framework. Unlike conventional keyword-based retrieval systems that primarily optimize lexical similarity, the proposed framework combines retrieval effectiveness with transparency, explainability, and user trust through a Mamdani fuzzy inference system and metadata-driven explanation generation.

The evaluation demonstrated that the proposed framework maintained competitive retrieval performance while providing additional human-centered capabilities that are not reflected by conventional IR metrics alone. The combination of quantitative benchmarking and qualitative user-centered evaluation therefore provides complementary evidence supporting the feasibility of transparent and explainable retrieval for OGD portals.

8.2 Summary of Research

This research addressed the challenge of improving dataset discoverability within the Swiss OGD ecosystem by developing a HCIR framework that combines conventional IR techniques with fuzzy logic-based multi-criteria ranking and explainable retrieval mechanisms. The work was motivated by the observation that traditional keyword-based retrieval approaches primarily emphasize lexical similarity while providing limited support for transparency, exploratory search, and user understanding of ranking decisions.

Following the DSR methodology, a complete retrieval framework was designed, implemented, and evaluated. The developed system integrates a custom BM25 retrieval component, a Mamdani fuzzy inference engine, metadata-driven ranking criteria, and an explanation generation module into a unified retrieval pipeline. Rather than replacing conventional retrieval algorithms, the framework extends them by incorporating human-centered relevance criteria, including metadata completeness, dataset recency, organizational quality, and transparent explanation of ranking decisions.

The implemented prototype was integrated with the live CKAN API of the Swiss OGD portal, allowing users to retrieve real datasets while interacting with an explainable ranking interface. The framework also provides interactive controls that allow users to adjust ranking preferences and compare alternative retrieval approaches, thereby supporting exploratory information seeking and improving transparency during dataset discovery.

The proposed framework was evaluated using a mixed-method evaluation strategy consisting of quantitative IR benchmarking and qualitative user-centered assessment. The quantitative evaluation demonstrated that the proposed framework maintained competitive retrieval effectiveness compared with conventional retrieval approaches, while the qualitative evaluation showed positive participant perceptions regarding usability, transparency, explainability, and trust. Taken together, these findings indicate that fuzzy logic can successfully support HCIR by complementing traditional retrieval methods with interpretable multi-criteria ranking rather than replacing them.

Overall, the research demonstrates that transparent and explainable ranking mechanisms can improve the usability of OGD retrieval systems while maintaining practical retrieval effectiveness. The developed framework therefore contributes to the growing body of research on HCIR by showing how fuzzy logic can be applied to support more understandable and user-oriented dataset discovery.

8.3 Research Questions Revisited

This section revisits the research questions introduced in Chapter 1 and summarizes how they were addressed through the study.

8.3.1 RQ1: How can fuzzy logic be used to improve dataset retrieval within OGD portals?

The findings of this research demonstrate that fuzzy logic provides an effective mechanism for integrating multiple relevance criteria within a transparent HCIR framework. By combining keyword relevance, metadata completeness, dataset recency, organizational quality, and explanation generation through a Mamdani fuzzy inference system, the proposed framework extends conventional keyword-based retrieval beyond lexical similarity alone. The quantitative evaluation demonstrated competitive retrieval effectiveness, while the user-centered evaluation showed that participants valued the transparency and interpretability provided by the fuzzy ranking process. Together, these findings indicate that fuzzy logic can effectively enhance OGD retrieval by supporting interpretable multi-criteria decision making while maintaining practical retrieval performance.

8.3.2 RQ2: How can multiple metadata-based relevance criteria be integrated into a unified and interpretable ranking mechanism?

This research addressed the integration of multiple relevance criteria through a Mamdani fuzzy inference engine that combines metadata completeness, keyword relevance, recency, and organizational quality using linguistic variables and expert-defined inference rules. Unlike traditional weighted-sum approaches, fuzzy inference naturally models gradual transitions between different relevance levels while preserving interpretability. The developed explanation mechanism further enables users to understand how individual metadata attributes influence ranking decisions, thereby supporting transparent and understandable retrieval.

8.3.3 RQ3: To what extent does a fuzzy logic-based retrieval system support HCIR principles?

The evaluation demonstrates that the proposed framework successfully supports the primary objectives of HCIR by combining competitive retrieval effectiveness with transparency, explainability, and user trust. Although conventional retrieval approaches achieved higher values for several traditional IR metrics, participants consistently reported positive perceptions regarding usability, explanation quality, and confidence in the retrieval results. These findings suggest that the principal contribution of the framework lies in improving user understanding of ranking decisions while maintaining practical retrieval effectiveness.

8.3.4 RQ4: How effective are metadata-driven explanations in improving user understanding and trust in ranking decisions?

The qualitative evaluation indicates that metadata-driven explanations substantially improved participant understanding of retrieval decisions. Users consistently reported that explanation messages, relevance scores, and metadata summaries enabled them to interpret why datasets appeared in particular ranking positions and increased their confidence in the retrieval results. Although explanation quality ultimately depends on the availability and consistency of metadata, the findings demonstrate that interpretable explanation mechanisms represent an important component of HCIR for OGD portals.

8.4 Research Contributions

This research contributes to the field of HCIR by demonstrating how fuzzy logic can be integrated with conventional IR techniques to improve the discoverability and interpretability of Swiss OGD. The principal contributions of this thesis are summarized as follows.

1. A HCIR framework for Open Government Data.

This thesis proposes a HCIR framework that extends traditional keyword-based retrieval by integrating multiple metadata-based relevance criteria within an interpretable fuzzy inference process. Unlike conventional retrieval approaches that primarily emphasize lexical similarity, the proposed framework explicitly considers transparency, explainability, metadata quality, and user understanding during ranking.

2. A custom fuzzy multi-criteria ranking mechanism.

A Mamdani fuzzy inference engine was designed and implemented to combine keyword relevance, metadata completeness, dataset recency, and organizational quality into a unified ranking score. The rule-based design allows gradual reasoning under uncertainty while maintaining full interpretability of ranking decisions.

3. An explainable retrieval interface for Swiss Open Government Data.

The developed prototype integrates metadata-driven explanation generation directly into the retrieval interface. Rather than presenting opaque ranking scores, the system explains retrieval decisions using interpretable factors such as keyword matching, metadata completeness, publication recency, and dataset quality, thereby supporting user understanding and trust.

4. Integration with the live Swiss OGD portal.

Unlike many conceptual HCIR studies, the proposed framework was implemented as a functional prototype connected to the live CKAN API of the Swiss OGD portal. The system retrieves real datasets and demonstrates the practical feasibility of transparent and explainable retrieval within an operational OGD environment.

5. Comprehensive evaluation of technical and human-centered performance.

The proposed framework was evaluated using a mixed-method strategy that combined quantitative IR benchmarking with qualitative user-centered assessment. The evaluation demonstrated that the framework maintains competitive retrieval effectiveness while providing additional benefits in terms of transparency, explainability, and user trust, illustrating the value of combining technical and human-centered evaluation within DSR.

Overall, the research demonstrates that fuzzy logic can complement conventional IR by providing an interpretable multi-criteria ranking process that supports HCIR objectives without requiring the replacement of existing retrieval infrastructures.

8.5 Theoretical Implications

Beyond the implementation of a functional retrieval prototype, this research contributes to the theoretical understanding of HCIR by demonstrating that retrieval effectiveness alone is insufficient for evaluating exploratory OGD search systems. The findings indicate that transparency, explainability, and user trust represent complementary dimensions of retrieval quality that should be considered alongside traditional IR performance metrics.

The proposed framework further demonstrates that fuzzy logic provides a practical and interpretable mechanism for modelling the uncertainty that naturally occurs during exploratory information seeking. Rather than relying exclusively on precise numerical relevance scores, the Mamdani fuzzy inference system enables gradual reasoning using linguistic concepts such as *low*, *medium*, and *high* relevance. This approach aligns more closely with the way users evaluate the usefulness of datasets during exploratory search, where relevance is often subjective rather than binary.

The evaluation also highlights the importance of integrating technical and human-centered evaluation methods within DSR. While quantitative IR metrics provide objective measures of retrieval performance, the qualitative user study demonstrates that user perceptions of transparency, explanation quality, and trust represent equally important indicators of system effectiveness. The combination of both evaluation approaches therefore provides a more comprehensive understanding of retrieval quality than either method alone.

Finally, this research demonstrates that explainability should be considered an integral component of retrieval system design rather than an optional interface feature. By explicitly communicating the factors influencing ranking decisions, the proposed framework supports informed decision making and enables users to better understand and trust the retrieval process. These findings contribute to the growing body of research advocating transparent and user-centered approaches to IR and decision support systems [40, 26, 14, 9].

8.6 Practical Implications

The outcomes of this research demonstrate that HCIR can provide practical benefits for the design and operation of OGD portals by improving the transparency and interpretability of dataset retrieval. While conventional keyword-based retrieval systems remain effective for identifying relevant datasets, the proposed framework illustrates how explainable multi-criteria ranking can improve the overall search experience without replacing existing retrieval infrastructures.

For OGD providers, the proposed framework offers a practical approach for enhancing dataset discoverability while preserving compatibility with existing CKAN-based search platforms. Since the fuzzy inference engine operates as an additional ranking layer, the framework can be incorporated into existing retrieval systems with relatively limited architectural changes. The explanation mechanism further enables users to understand why datasets appear in particular ranking positions, thereby increasing confidence in retrieval results and encouraging more effective exploration of public datasets.

The findings also have practical implications for government organizations responsible for publishing OGD. The evaluation highlights the importance of high-quality metadata, including accurate descriptions, meaningful keywords, complete resource information, and up-to-date publication dates. Improvements in metadata quality not only enhance conventional retrieval methods but also strengthen explanation generation and multi-criteria ranking within human-centered retrieval systems.

From a broader perspective, the research demonstrates that retrieval systems should be evaluated using both technical performance measures and user-centered criteria. The combination of quantitative IR benchmarking with qualitative user evaluation provides a more comprehensive assessment of retrieval quality than either approach alone. This evaluation strategy may be valuable for future HCIR research and for the development of transparent decision-support systems in other application domains.

Finally, the proposed framework provides a practical foundation for future retrieval systems that seek to balance retrieval effectiveness with transparency, explainability, and user trust. These characteristics are becoming increasingly important as public-sector information systems continue to evolve toward more user-oriented and explainable digital

services [14, 40, 9].

8.7 Limitations

Although the proposed HCIR framework achieved the objectives of this research, the findings should be interpreted within the scope of the evaluation conducted in this study.

First, the quantitative evaluation was performed using a benchmark developed for the Swiss OGD portal. While the benchmark provided a consistent basis for comparing different retrieval approaches, the relevance judgments were derived from metadata-based annotations and subsequently reviewed manually. Consequently, the reported IR metrics primarily evaluate ranking effectiveness with respect to lexical and metadata similarity and may not fully capture human-centered characteristics such as transparency, explainability, and user understanding.

Second, the qualitative evaluation involved ten Master’s students who participated in a controlled user study. Although this participant group was appropriate for evaluating a research prototype and provided valuable insights into user interaction with the proposed framework, the findings should not be interpreted as representing the broader population of OGD users. Additional studies involving larger and more diverse participant groups would further strengthen the external validity of the results.

Third, the framework was evaluated exclusively using datasets retrieved from the Swiss OGD portal. While this provided a realistic application environment, differences in metadata quality, portal architecture, and retrieval infrastructure may influence the performance of the proposed approach when applied to other OGD platforms.

These limitations do not diminish the overall contribution of the research. Instead, they define the scope within which the reported findings should be interpreted and identify opportunities for extending the proposed framework in future studies.

8.8 Future Work

The findings of this research provide several opportunities for extending the proposed HCIR framework and further advancing research on transparent and explainable retrieval for OGD.

A natural direction for future work is the evaluation of the proposed framework with a larger and more diverse participant population. The current user study provided valuable formative insights into usability, transparency, and trust; however, involving government agencies, policy makers, researchers, journalists, and citizens would provide a broader understanding of user requirements and improve the generalizability of the findings. Larger participant groups would also enable more comprehensive statistical analyses of user behaviour and system acceptance.

Another important direction involves evaluating the framework across multiple OGD portals. The present study focused exclusively on the Swiss OGD portal because of its comprehensive metadata and open CKAN-based infrastructure. Applying the framework to additional national and international OGD portals would allow the robustness of the proposed ranking methodology to be assessed across different metadata standards, retrieval infrastructures, and application domains [25].

Future research may also investigate the integration of modern IR techniques with the proposed HCIR framework. Dense retrieval models, learning-to-rank approaches, and Large Language Model (LLM)-assisted retrieval have demonstrated promising retrieval performance in recent years. Rather than replacing the proposed fuzzy inference system, these techniques could be incorporated as complementary retrieval components while preserving the transparency and explainability that represent the principal strengths of the proposed framework.

The fuzzy inference engine itself offers several opportunities for further development. Future studies could investigate adaptive optimization of membership functions and fuzzy rules using user interaction data or machine learning techniques while maintaining the interpretability of the ranking process. Such an approach could improve retrieval performance without sacrificing the transparency that distinguishes fuzzy rule-based reasoning from many black-box learning methods.

Another promising direction involves introducing personalized HCIR. The current framework applies a common ranking strategy for all users, whereas future systems could adapt ranking preferences according to user expertise, search objectives, domain knowledge, or previous search behaviour. Combining personalization with transparent explanation mechanisms would allow users to understand not only why datasets were retrieved but also how their individual preferences influenced ranking decisions.

Finally, future work could further enhance the explanation component by generating richer natural-language explanations that combine metadata analysis, fuzzy reasoning, and user context. Such explanations could improve user understanding while supporting greater confidence in retrieval decisions, particularly during exploratory information seeking and decision-support tasks [40, 26].

Overall, the proposed framework establishes a strong foundation for future HCIR research by demonstrating that transparent, explainable, and user-oriented retrieval can successfully complement conventional IR techniques within OGD environments.

8.9 Final Remarks

This thesis set out to investigate how HCIR can improve access to Swiss OGD through the integration of fuzzy logic-based multi-criteria ranking and explainable retrieval mechanisms. The research has demonstrated that effective IR is not solely determined by

ranking accuracy, but also by the extent to which users can understand, interpret, and trust the retrieval process.

The proposed framework combines conventional IR techniques with an interpretable Mamdani fuzzy inference system and metadata-driven explanation generation to support transparent and user-centered dataset discovery. Through its implementation and evaluation, the research demonstrates that fuzzy logic can successfully complement traditional keyword-based retrieval by incorporating multiple human-centered relevance criteria while maintaining competitive retrieval effectiveness.

Beyond the technical implementation, this research contributes to the broader objective of making OGD more accessible, understandable, and usable. By enabling users to better understand why datasets are retrieved and how ranking decisions are formed, the proposed framework supports more informed exploration of public information and encourages greater confidence in OGD systems.

Although opportunities remain for extending the framework through larger-scale evaluations, broader deployment, and integration with emerging retrieval technologies, the research establishes a practical foundation for future work on transparent and explainable HCIR.

In conclusion, this thesis demonstrates that HCIR can enhance OGD discovery by integrating interpretable fuzzy multi-criteria ranking with transparent explanation mechanisms. Rather than aiming to replace conventional IR techniques, the proposed framework complements existing lexical retrieval approaches by supporting exploratory search, improving user understanding of ranking decisions, and promoting greater transparency in dataset discovery. Although the evaluation was conducted on a limited corpus and participant sample, the results indicate that transparent ranking approaches can provide meaningful benefits beyond conventional retrieval effectiveness metrics. The proposed framework therefore establishes a practical and reproducible foundation for future research on explainable and human-centered retrieval systems within OGD environments. The findings of this research contribute to the growing body of work on transparent and explainable IR by demonstrating that effective retrieval systems should be evaluated not only by their ranking performance but also by their ability to support human understanding, trust, and informed decision-making.

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Appendix A

Evaluation Questionnaire

Participants were asked to indicate their level of agreement with the following statements using a five-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5).

1. The system was easy to use.
2. The search interface was intuitive.
3. The retrieved datasets were relevant to my search query.
4. The ranking of datasets appeared reasonable.
5. The explanations helped me understand ranking decisions.
6. The system appeared transparent.
7. The explanations increased my confidence in the results.
8. The system was useful for dataset discovery.
9. I would use this system again.
10. Overall, I was satisfied with the system.

Appendix B

Example Fuzzy Rules

The following examples illustrate the fuzzy inference rules used within the proposed retrieval framework.

1. IF BM25 Similarity is High AND Metadata Completeness is High AND Dataset Recency is High AND Resource Availability is High THEN Relevance is Very High.
2. IF BM25 Similarity is High AND Metadata Completeness is Medium THEN Relevance is High.
3. IF BM25 Similarity is Medium AND Metadata Completeness is High THEN Relevance is High.
4. IF BM25 Similarity is Medium AND Dataset Recency is High THEN Relevance is Medium.
5. IF BM25 Similarity is Low THEN Relevance is Low.

B.1 Calibrated Membership Function Parameters

The Mamdani Fuzzy Inference System employs data-calibrated membership functions derived from the statistical characteristics of the evaluation corpus. Rather than selecting linguistic boundaries manually, the calibration process determines breakpoints from the observed distributions of metadata completeness, dataset recency, and resource availability. Similarity and the output relevance variable use predefined membership functions to preserve a stable interpretation of textual relevance across retrieval tasks. Table [B.1](#) summarises the calibrated membership functions used by the prototype.

Table B.1: Membership functions used by the Mamdani Fuzzy Inference System.

Variable	Linguistic Term	Type	Parameters
Similarity	Not Relevant	Triangular	(0.000, 0.000, 0.150)
	Somewhat Relevant	Triangular	(0.100, 0.250, 0.400)
	Relevant	Triangular	(0.300, 0.500, 0.700)
	Highly Relevant	Triangular	(0.600, 0.780, 0.920)
	Exact Match	Triangular	(0.850, 0.950, 1.000)
Metadata Completeness	Low	Triangular	(0.000, 0.000, 0.564)
	Partial	Triangular	(0.564, 0.649, 0.814)
	Medium	Trapezoidal	(0.649, 0.814, 0.819, 0.865)
	High	Trapezoidal	(0.814, 0.819, 0.865, 1.000)
	Complete	Triangular	(0.865, 1.000, 1.000)
Dataset Recency	Very Recent	Triangular	(1.000, 1.000, 1.000)
	Recent	Trapezoidal	(1.000, 1.000, 1.000, 49.500)
	Moderate	Trapezoidal	(1.000, 49.500, 98.000, 98.000)
	Old	Trapezoidal	(49.500, 98.000, 98.000, 98.000)
	Very Old	Triangular	(98.000, 98.000, 98.000)
Resource Availability	Minimal	Triangular	(0.000, 0.000, 1.000)
	Limited	Triangular	(0.000, 1.000, 2.000)
	Moderate	Triangular	(1.000, 2.000, 2.000)
	Rich	Trapezoidal	(2.000, 2.000, 4.000, 4.550)
	Comprehensive	Triangular	(4.000, 4.550, 6.000)
Overall Relevance	Very Low	Triangular	(0.000, 0.000, 0.250)
	Low	Triangular	(0.100, 0.280, 0.450)
	Moderate	Triangular	(0.350, 0.500, 0.650)
	Good	Triangular	(0.550, 0.720, 0.880)
	Excellent	Triangular	(0.800, 0.920, 1.000)

Appendix C

Example Retrieval Results

This appendix presents examples of retrieval outputs generated by the proposed framework. Each result includes the dataset title, relevance score, metadata summary, and generated explanation.

Example:

Query: Transportation

Dataset: Public Transport Stops in Switzerland

Relevance Score: 0.87

Explanation: The dataset achieved a high relevance score because it closely matched the query terms, contained highly complete metadata, included multiple accessible resources, and had been updated recently.